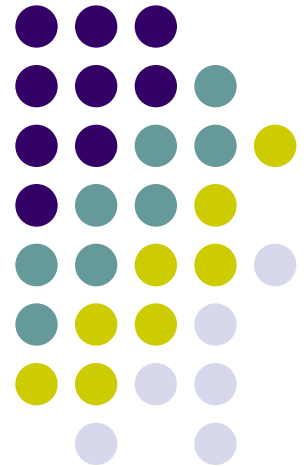
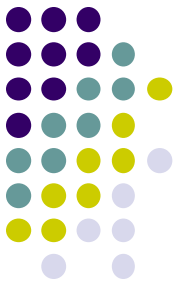


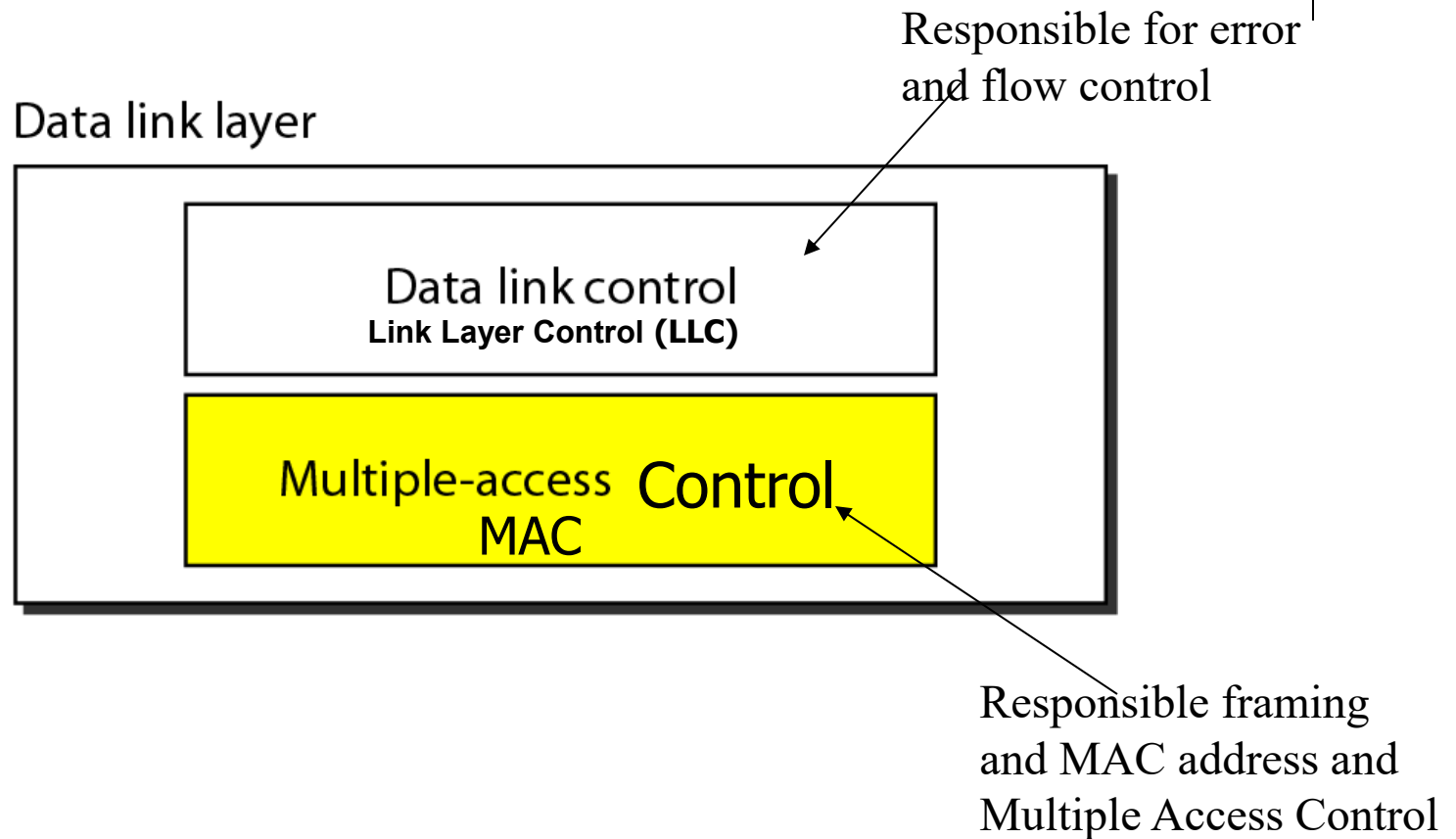
Chapter 3

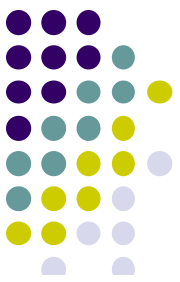
Multiple Access Technology





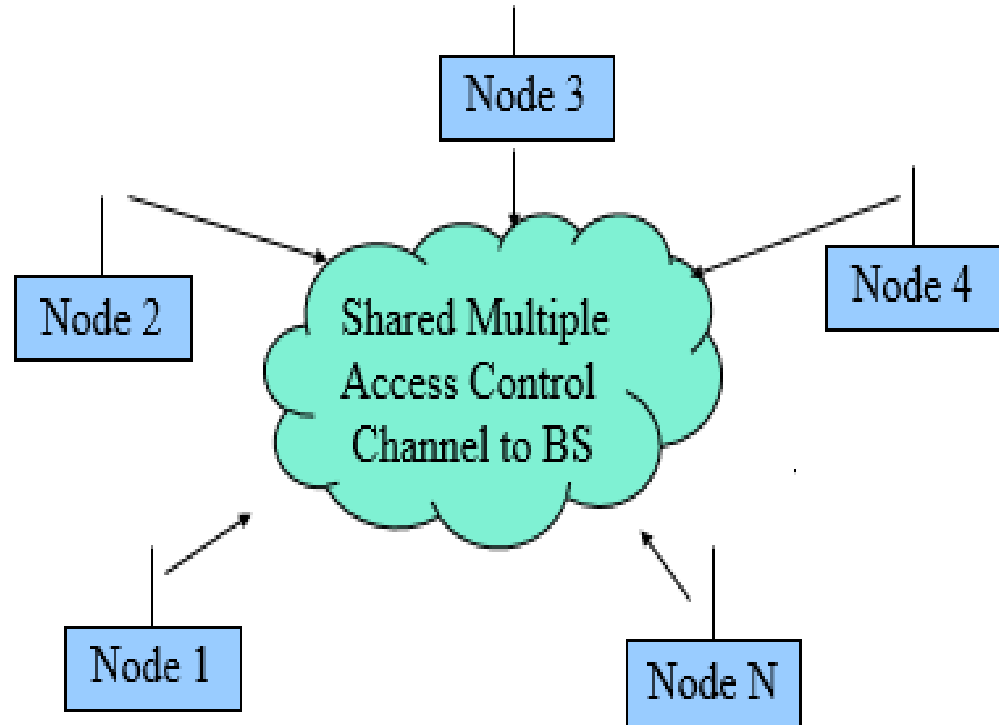
Data link layer divided into two functionality-oriented sublayers

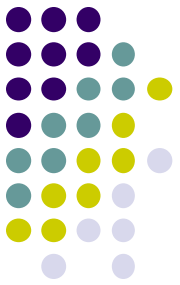




Introduction

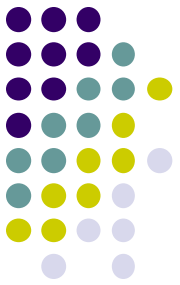
- Multiple access control channels
 - Each node is attached to a transmitter/receiver which communicates via a channel shared by other nodes
- Transmission from any node is received by other nodes



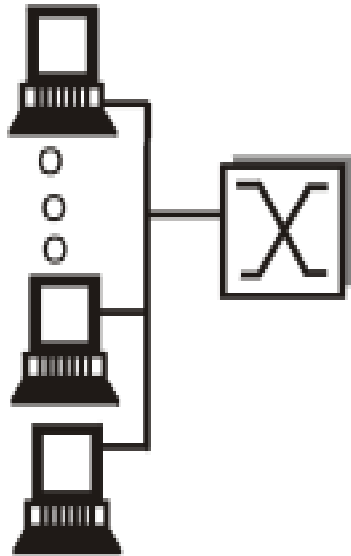


Multiple-access protocols

- When more than two nodes send at the same time , the transmitted frames collide.
- All collide frames are lost and the bandwidth of the broadcast channel will be wasted.
- We need multiple –access protocol to coordinate access to multipoint or broadcast link (Nodes or stations are connected to or use a common link)
- Multiple access protocols are needed in wire and wireless LANs and satellite networks



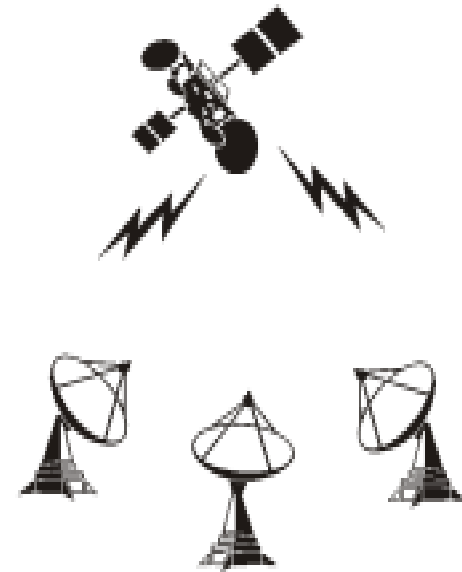
Multiple-access protocols... **cont'd**



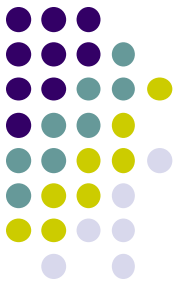
shared wire
(e.g. Ethernet)



shared wireless
(e.g. Wavelan)

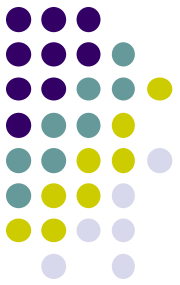


satellite



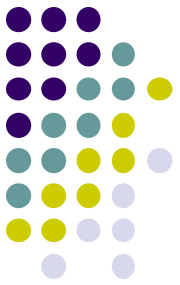
Multiple-access protocols... **Cont'd**

- *Problem of controlling the access to the medium is similar to the rules of speaking in an assembly*
 - Given everyone a chance to speak
 - Don't speak until you are spoken to
 - Don't monopolize the conversation
 - Raise your hand if you have a question
 - Don't interrupt when someone speaking



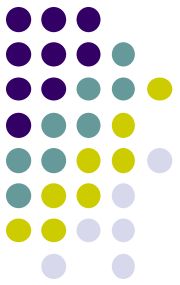
Multiple Access... Cont'd

- **Problem:** When two or more nodes transmit at the same time, their frames will collide and the link bandwidth is wasted during collision
 - How to coordinate the access of multiple sending/receiving nodes to the shared link???
 - **Solution:** We need a protocol to coordinate the transmission of the active nodes
- These protocols are called Medium or Multiple Access Control (MAC) Protocols belong to a sublayer of the data link layer called MAC (Medium Access Control)
- What is expected from Multiple Access Protocols:
- Main task is to **minimize collisions** in order to utilize the bandwidth by:
 - Determining when a station can use the link (medium)
 - what a station should do when the link is busy
 - what the station should do when it is involved in collision



Terms

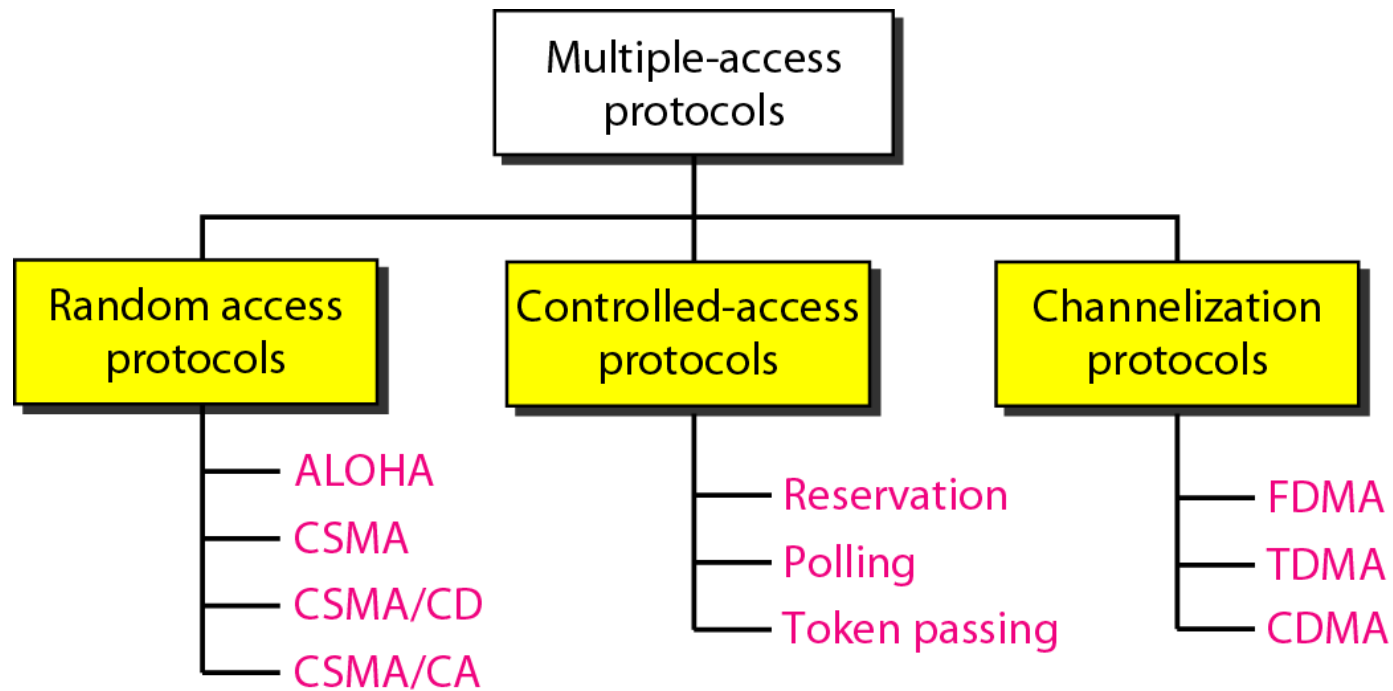
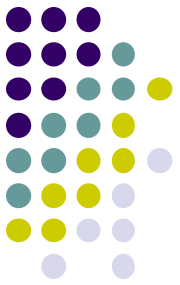
- **Throughput:** is the amount of data moved successfully from one place to another in a given time period.
- **propagation delay:** is the amount of time it takes for the head of the signal to travel from the sender to the receiver.

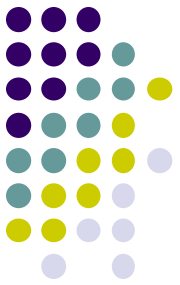


Multiple Access Technology

- Multiple access techniques are used to allow sharing of a finite amount of radio spectrum by many simultaneous users.
- The major multiple access techniques:
 - Frequency Division Multiple Access (FDMA)
 - Time Division Multiple Access (TDMA)
 - Code Division Multiple Access (CDMA)
 - Direct Sequence CDMA
 - Frequency Hopping CDMA
 - Space Division Multiple Access (SDMA)
- Contention based Protocols
 - ALOHA
 - CSMA CSMA/CD CSMA/CA

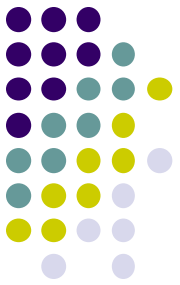
Taxonomy of multiple-access protocols





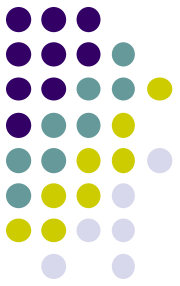
Channelization Protocol

- Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.
 - In this section, we discuss three channelization protocols.
 - FDMA
 - TDMA
 - CDMA



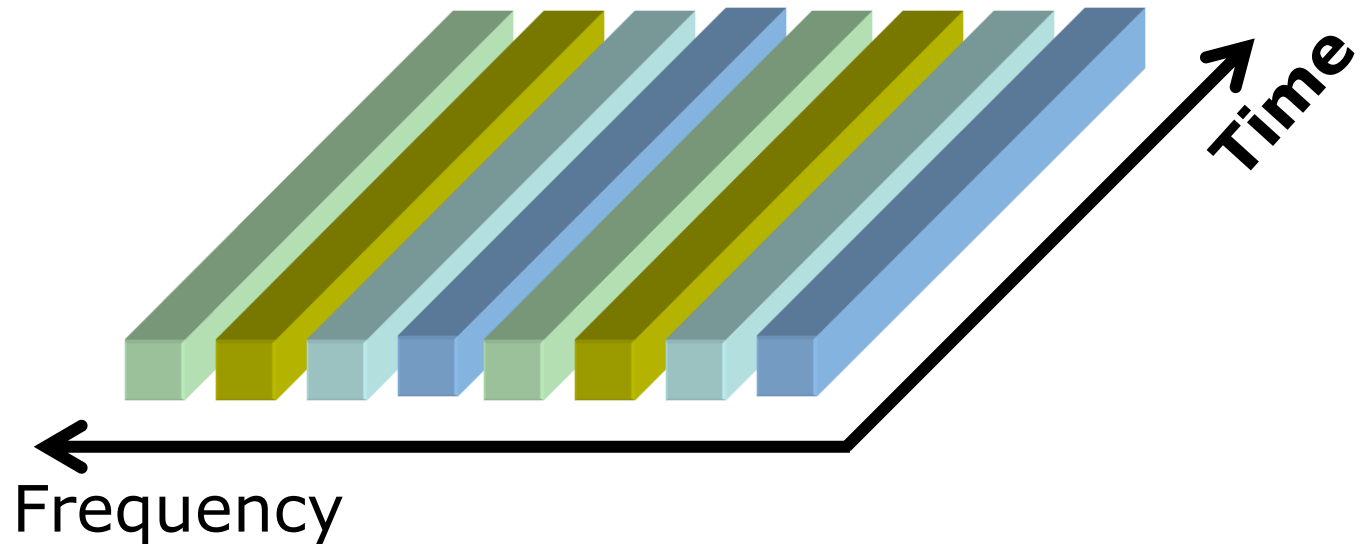
Frequency Division Multiple Access (FDMA)

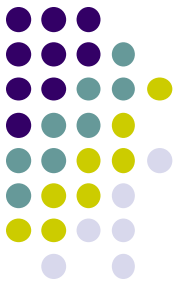
- The frequency band is divided into channels of equal bandwidth such that each conversation is carried on a different frequency.
- Best suited to analog mobile radio.
 - Single channel per carrier
 - All first generation systems use FDMA



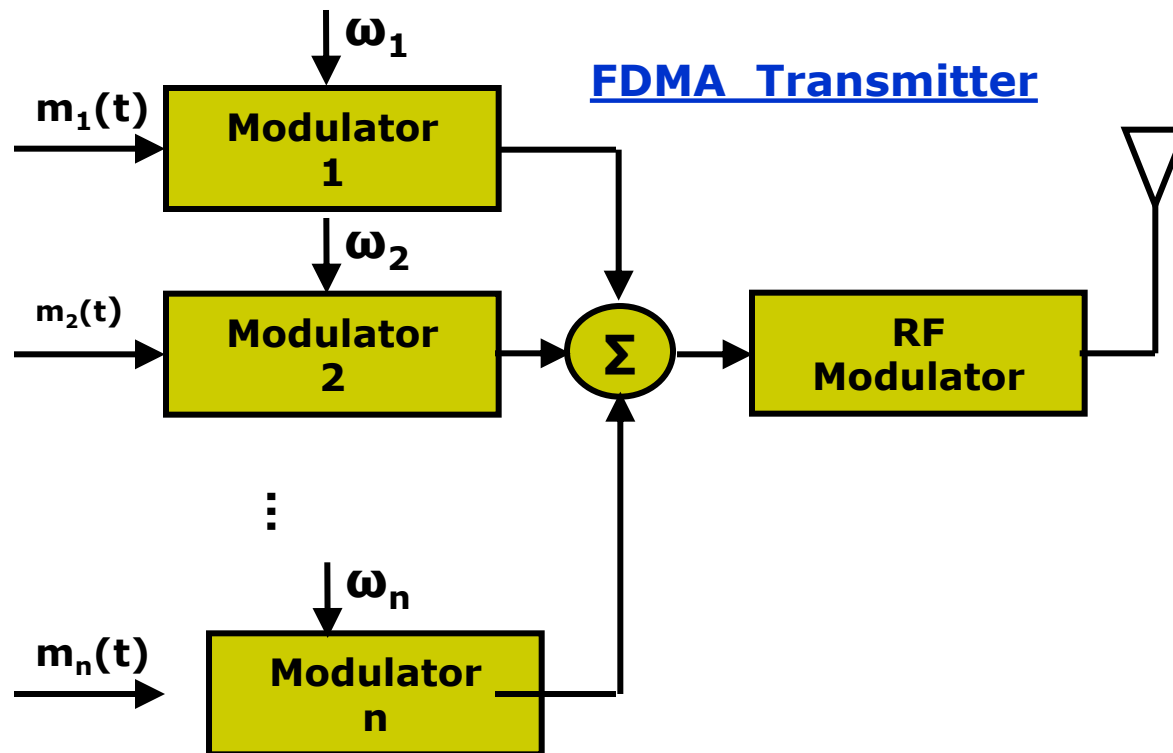
Channels of an FDMA system (schematic)

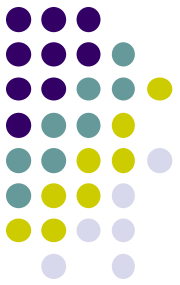
- FDMA assigns individual channels to individual users.
- During the period of the call, no other user can share the same channel.



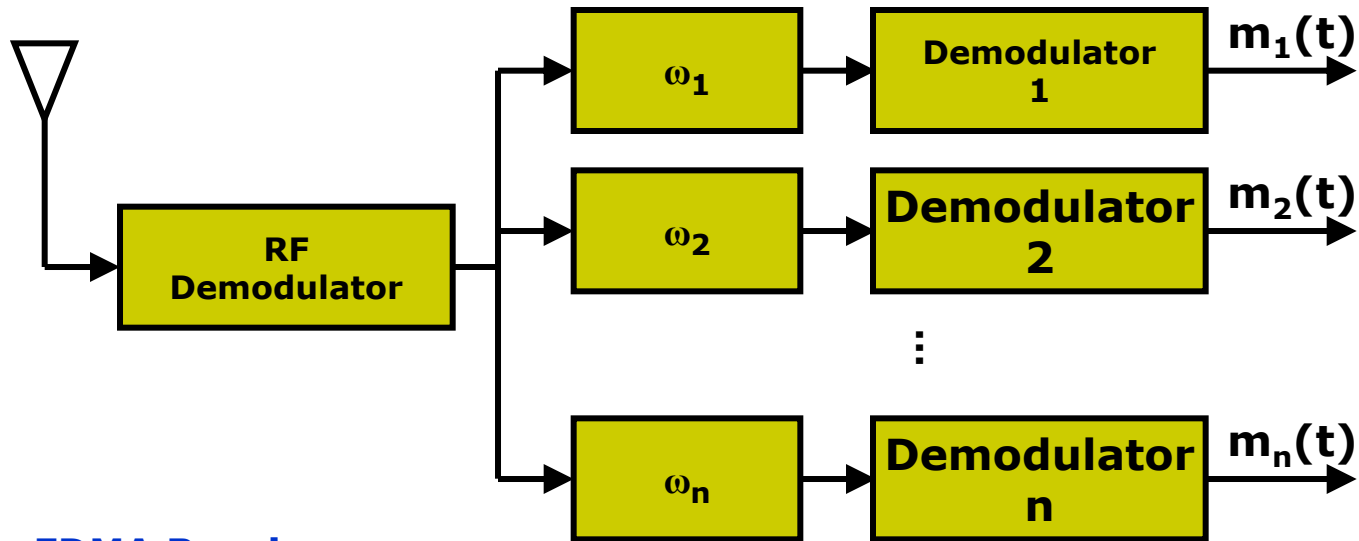


FDMA Transmitter

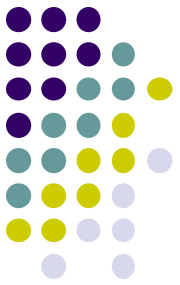




FDMA Receiver

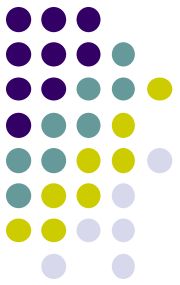


FDMA Receiver



Examples in Mobile Comm. System

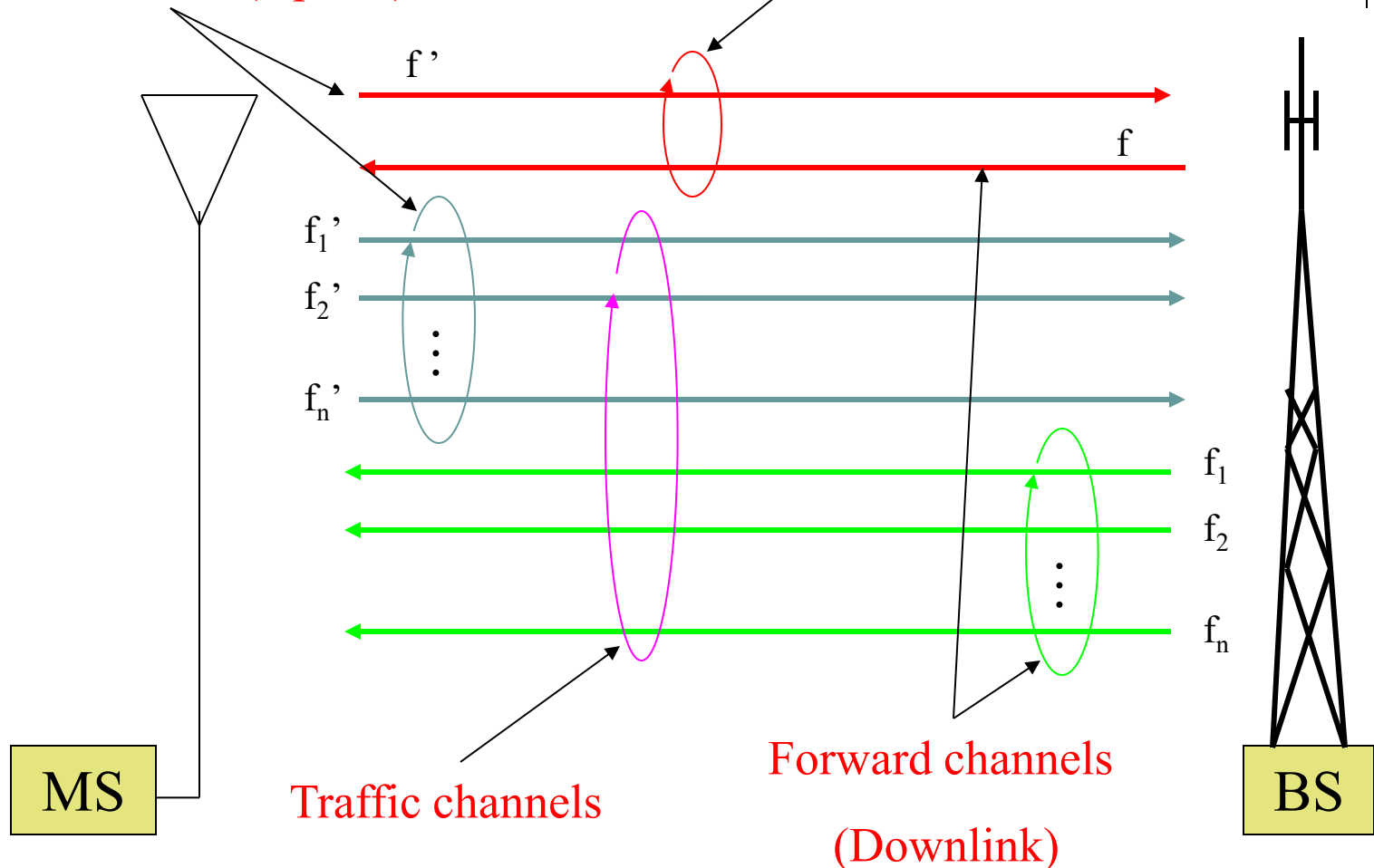
- Control channel
 - Forward (Downlink) control channel
 - Reverse (Uplink) control channel
- Traffic channel
 - Forward traffic (traffic or information) channel
 - Reverse traffic (traffic or information) channel



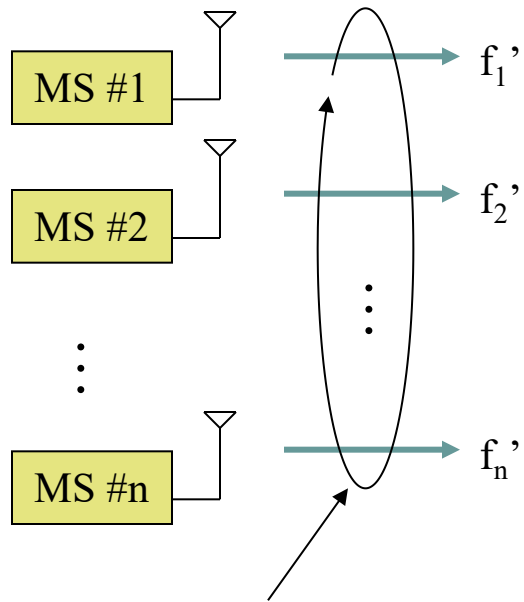
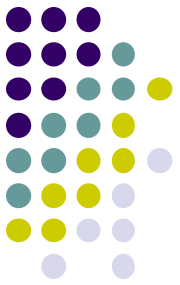
Types of Channels

Reverse channel (Uplink)

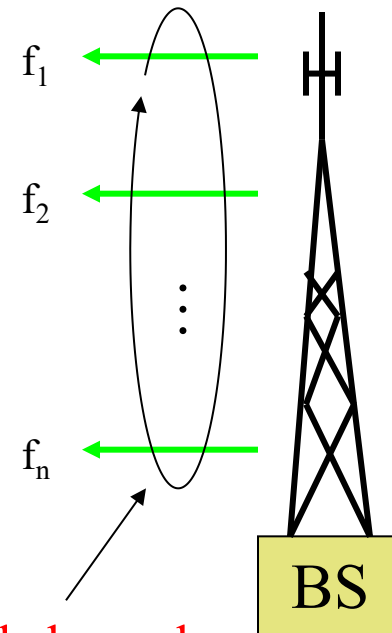
Control channels



FDMA

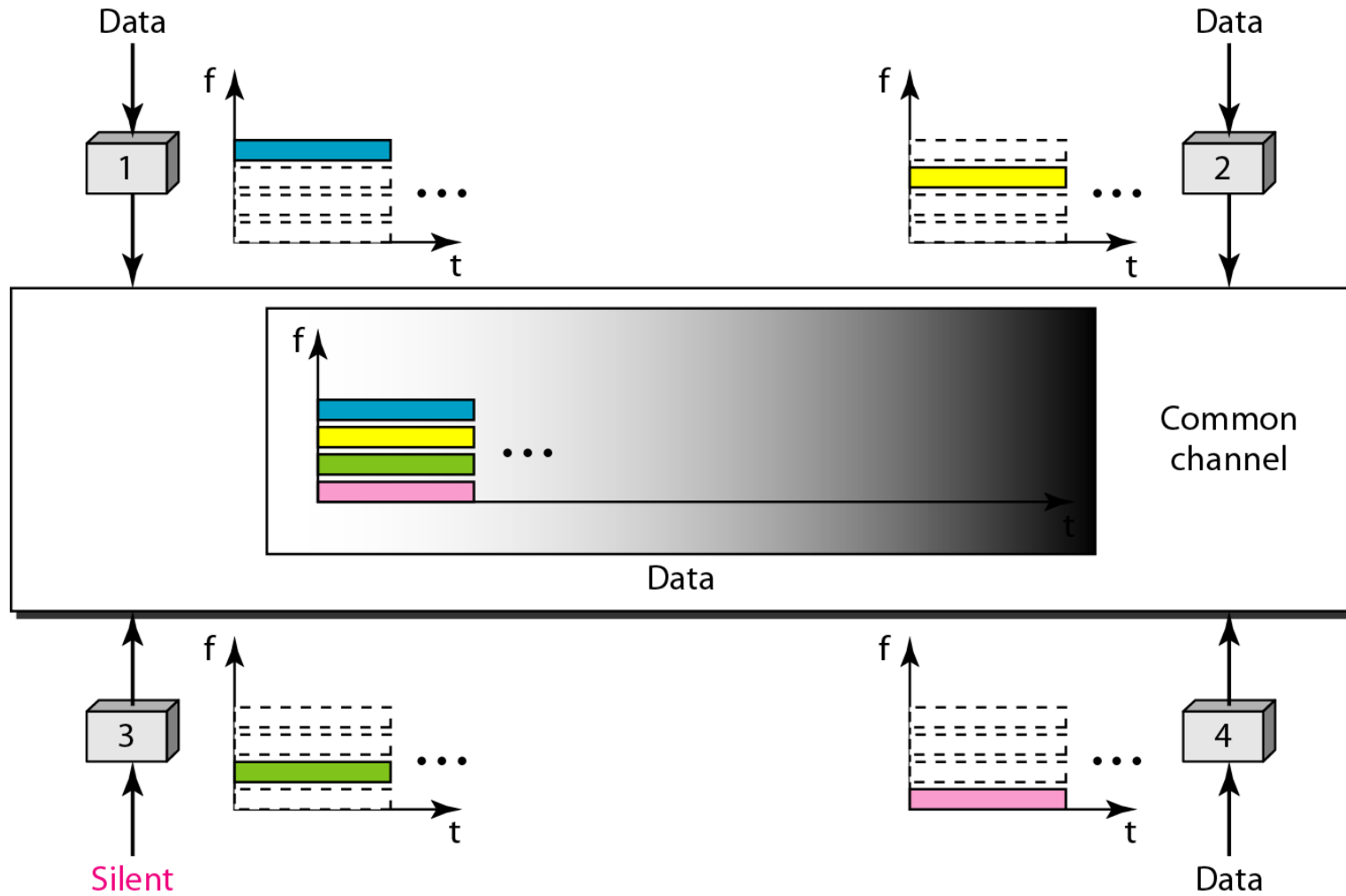


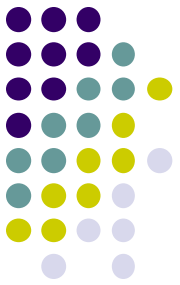
Reverse channels
(Uplink)



Forward channels
(Downlink)

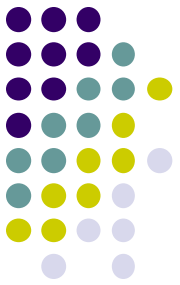
Frequency-division multiple access (FDMA)





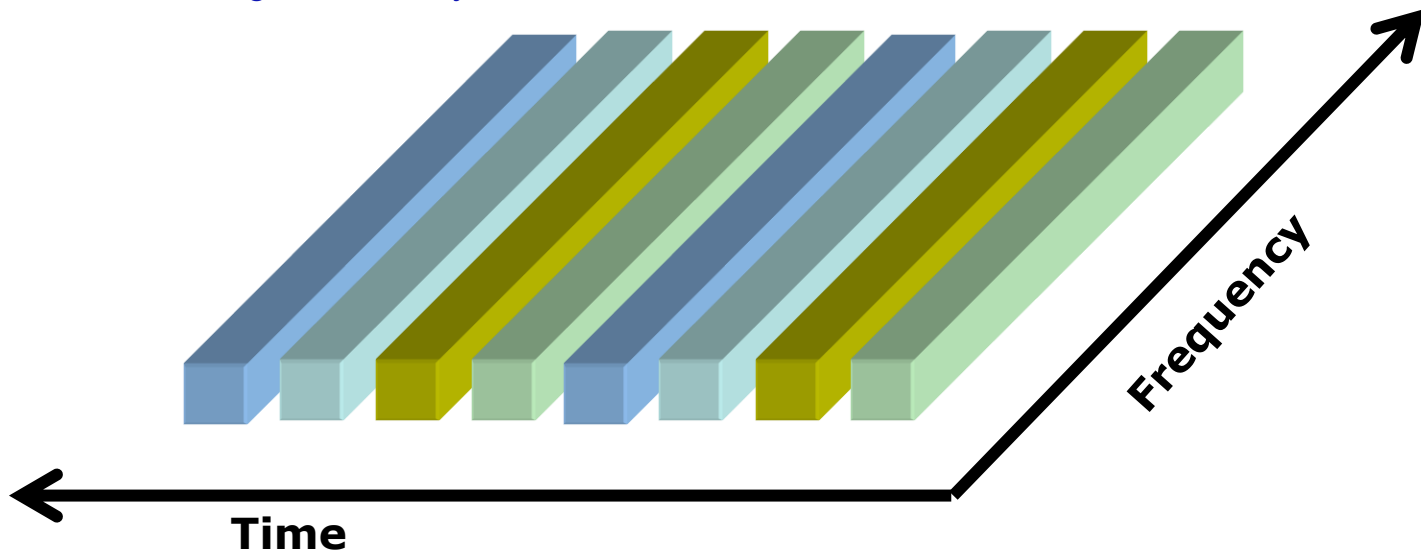
The features of FDMA

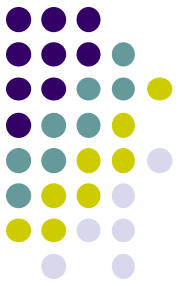
- The FDMA channel carries only one phone circuit at any time.
- An idle FDMA channel cannot be used by another user.
- FDMA is usually implemented in narrowband systems.
- The symbol time is large compared to the average delay spread, therefore, the amount of ISI is low, and little or no equalization is required.
- Tight filtering is required which increases the cost of both mobile terminal and a base station.
- Because of the nonlinearities, there is a problem of intermodulation interference, resulting in the adjacent-channel interference.
- To limit the adjacent-channel interference to other systems, a guard band at the edges of allocated spectrum is used.



Time Division Multiple Access (TDMA)

- TDMA systems divide the radio spectrum into time slots, and in each slot only one user is allowed to either transmit or receive.
- TDMA systems transmit data in a buffer-and burst method, thus the transmission for any user is noncontiguous.
 - Multiple channels per carrier
 - Most of second generation systems use TDMA

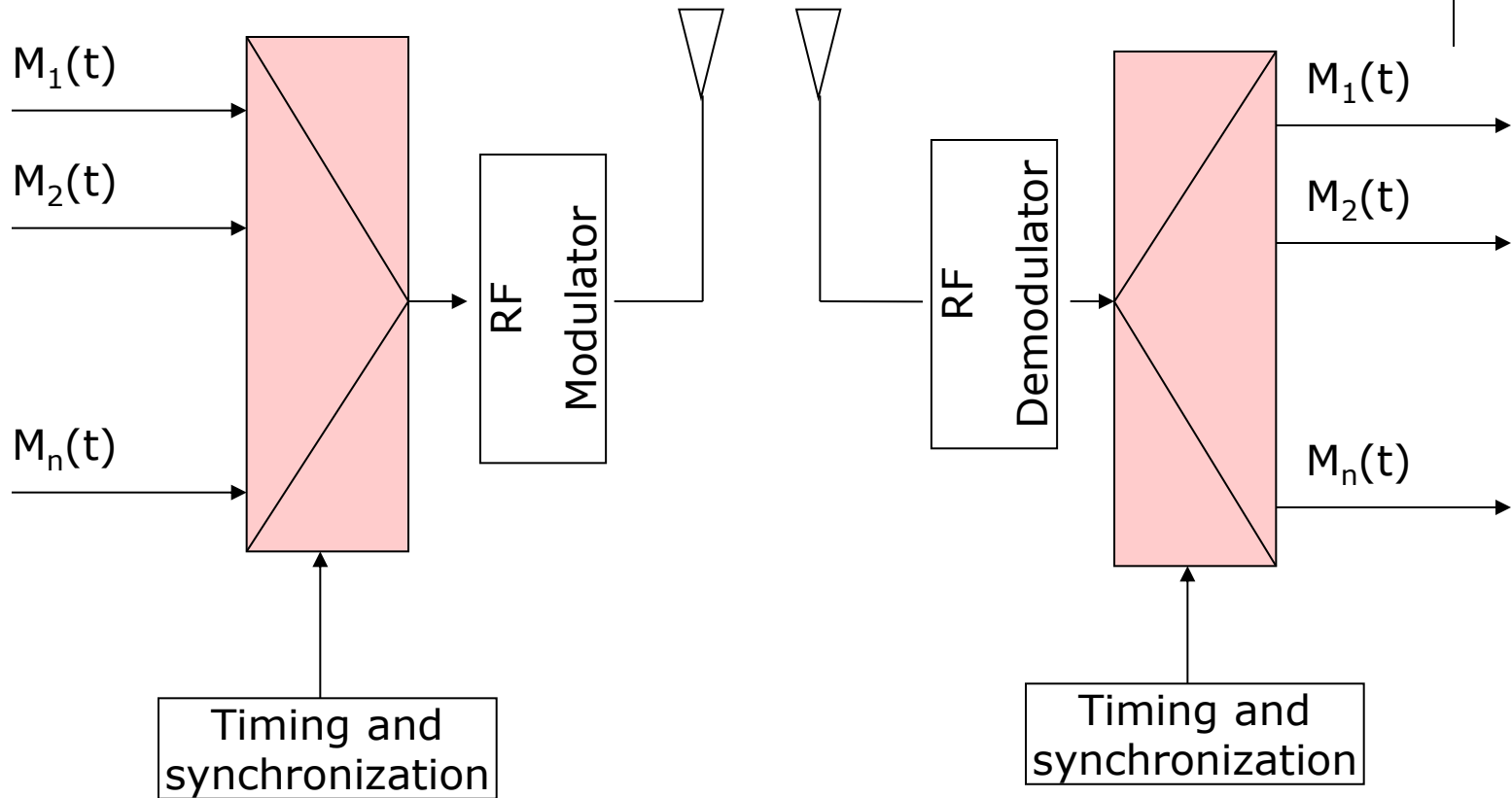




Time Division Multiple Access (TDMA)

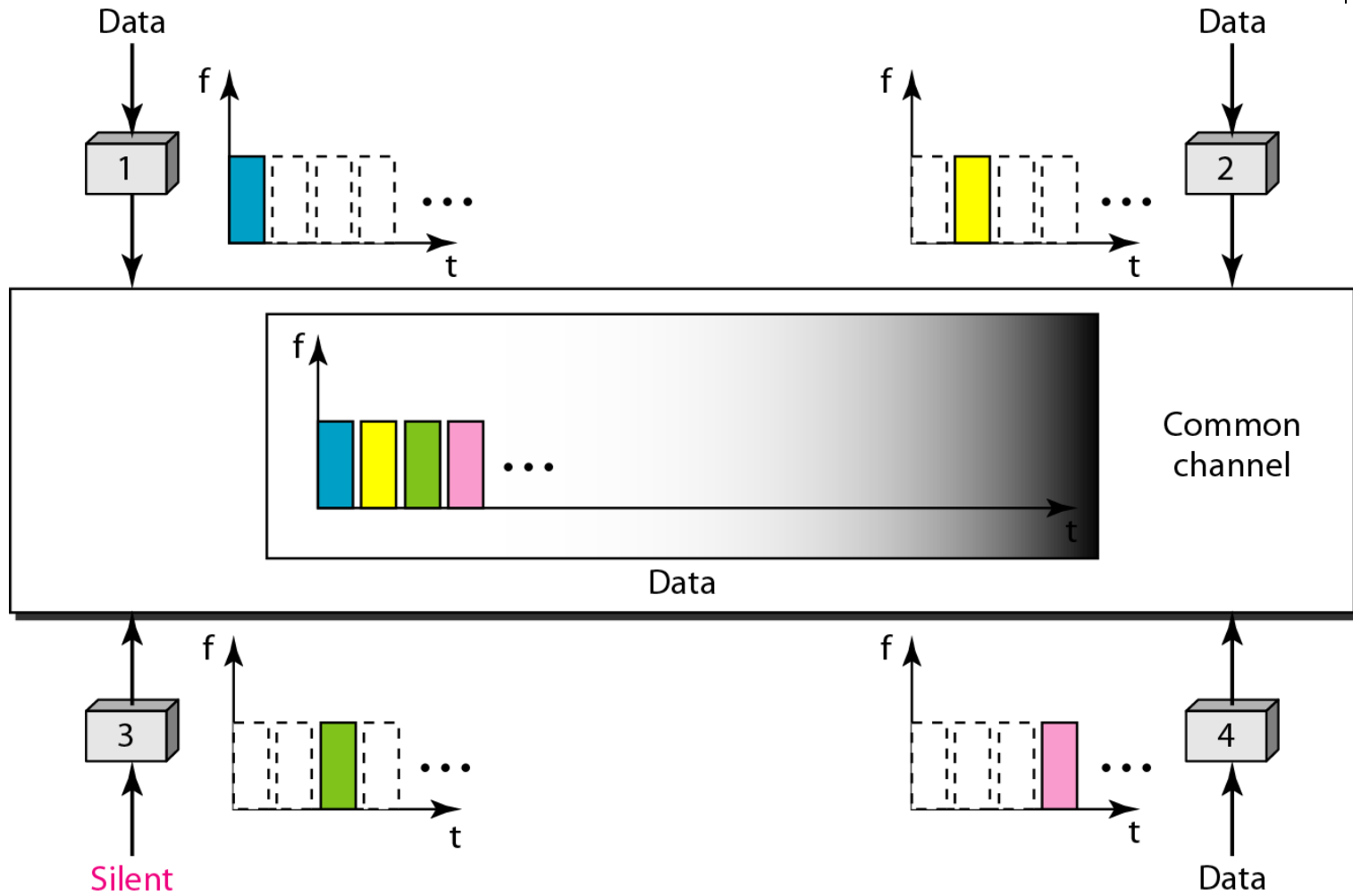
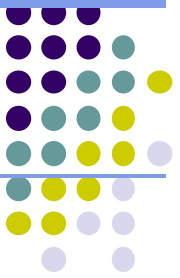
- The individual mobile stations are cyclically assigned a frequency for exclusive use only for the duration of a time slot.
- TDMA is a more expensive technique
 - needs a highly accurate synchronization between transmitter and receiver
- The TDMA technique is used in digital mobile radio systems.

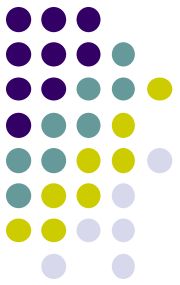
TDMA system



Simplified block diagram of a TDMA system

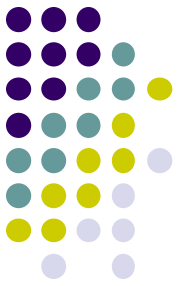
Time-division multiple access (TDMA)





The features of TDMA

- TDMA shares a single carrier frequency with several users, where each user makes use of nonoverlapping time slots.
- Data transmission for users of TDMA system occurs in bursts.
- Because of a discontinuous transmission, the handoff process is simpler for a mobile unit, as it can listen to the base stations during idle time slots.
- TDMA uses different time slots for transmission and the reception.
- Because of the transmission rates much higher than in FDMA, adaptive equalizers are usually required.
- The guard time between the slots is required.
- High synchronization overhead is required because of burst transmissions.



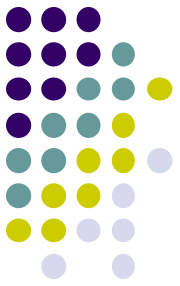
Activity

- What are the differences between TDM and TDMA?
- What are the differences between FDM and FDMA?

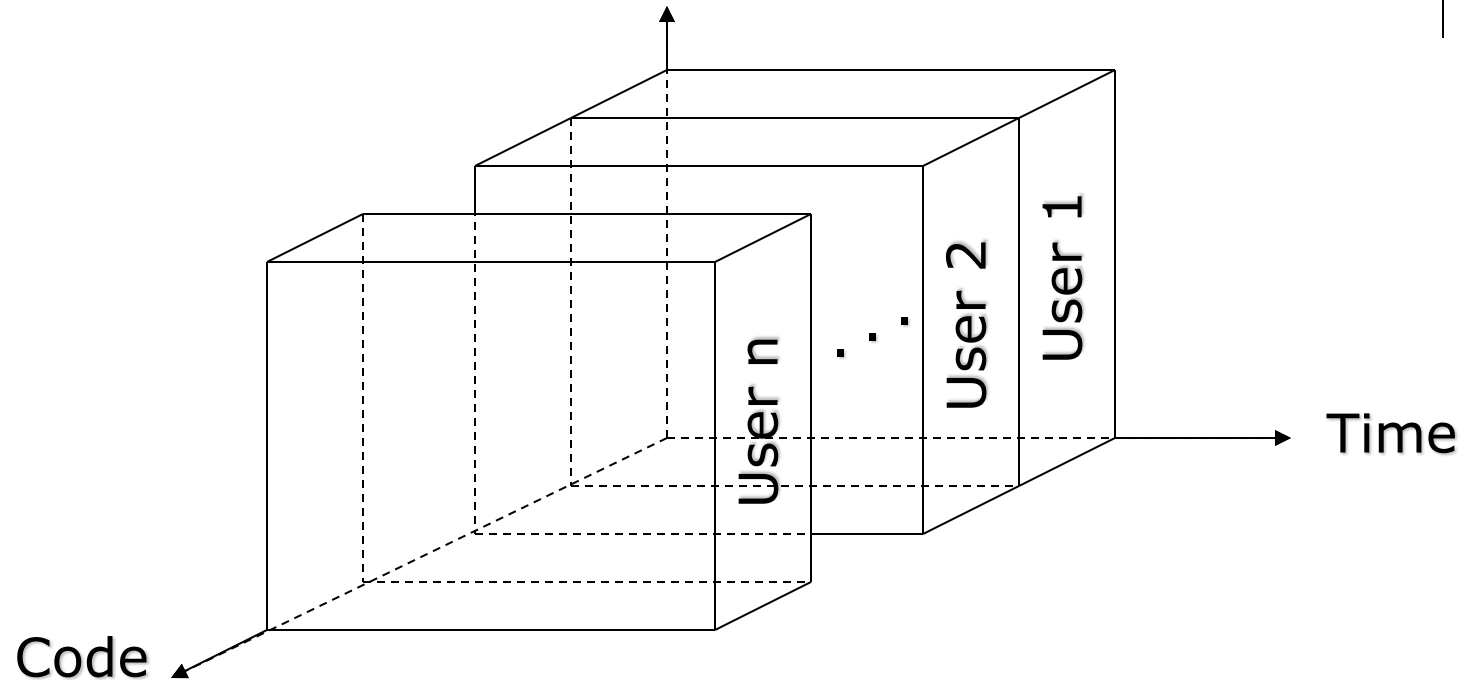


Code Division Multiple Access (CDMA)

- Systems with CDMA are broadband systems
- All the subscribers in a cell use the same frequency band simultaneously
- The subscribers are assigned orthogonal codes to separate the signals.
- The signal of one subscriber is spread spectrally over a multiple of its original bandwidth.

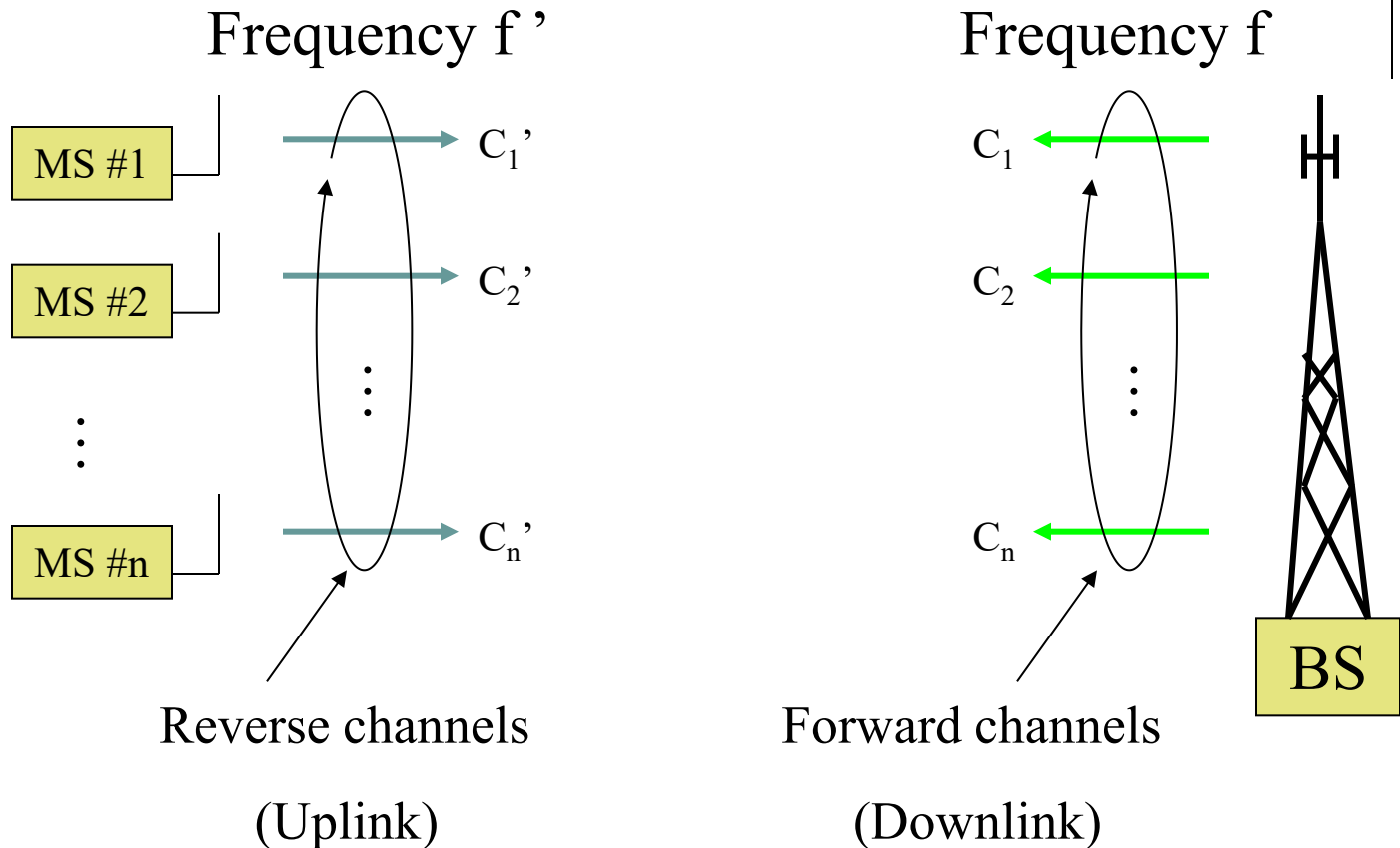
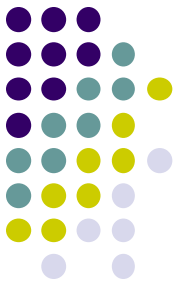


Code Division Multiple Access (CDMA)

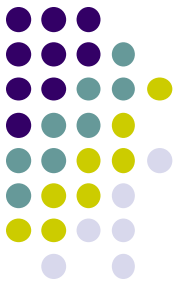


- Users share bandwidth by using code sequences that are orthogonal to each other
- Some second generation systems use CDMA
- Most of third generation systems use CDMA

Code Division Multiple Access (CDMA)



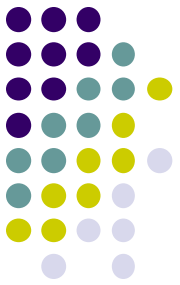
Note: $C_i' \times C_j' = 0$, i.e., C_i' and C_j' are orthogonal codes,
 $C_i \times C_j = 0$, i.e., C_i and C_j are orthogonal codes



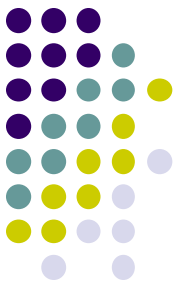
The features of CDMA

- Many users of a CDMA system share the same frequency.
- CDMA has a soft capacity limit.
- Multipath fading may be substantially reduced because the signal is spread over a large spectrum.
- Channel data rates are very high in CDMA systems.
- Since CDMA uses co-channel cells, it can use macroscopic spatial diversity to provide soft handoff.
- Self-jamming is a problem in CDMA system.
- The near-far problem occurs at a CDMA receiver if an undesired user has a high detected power as compared to the desired user.

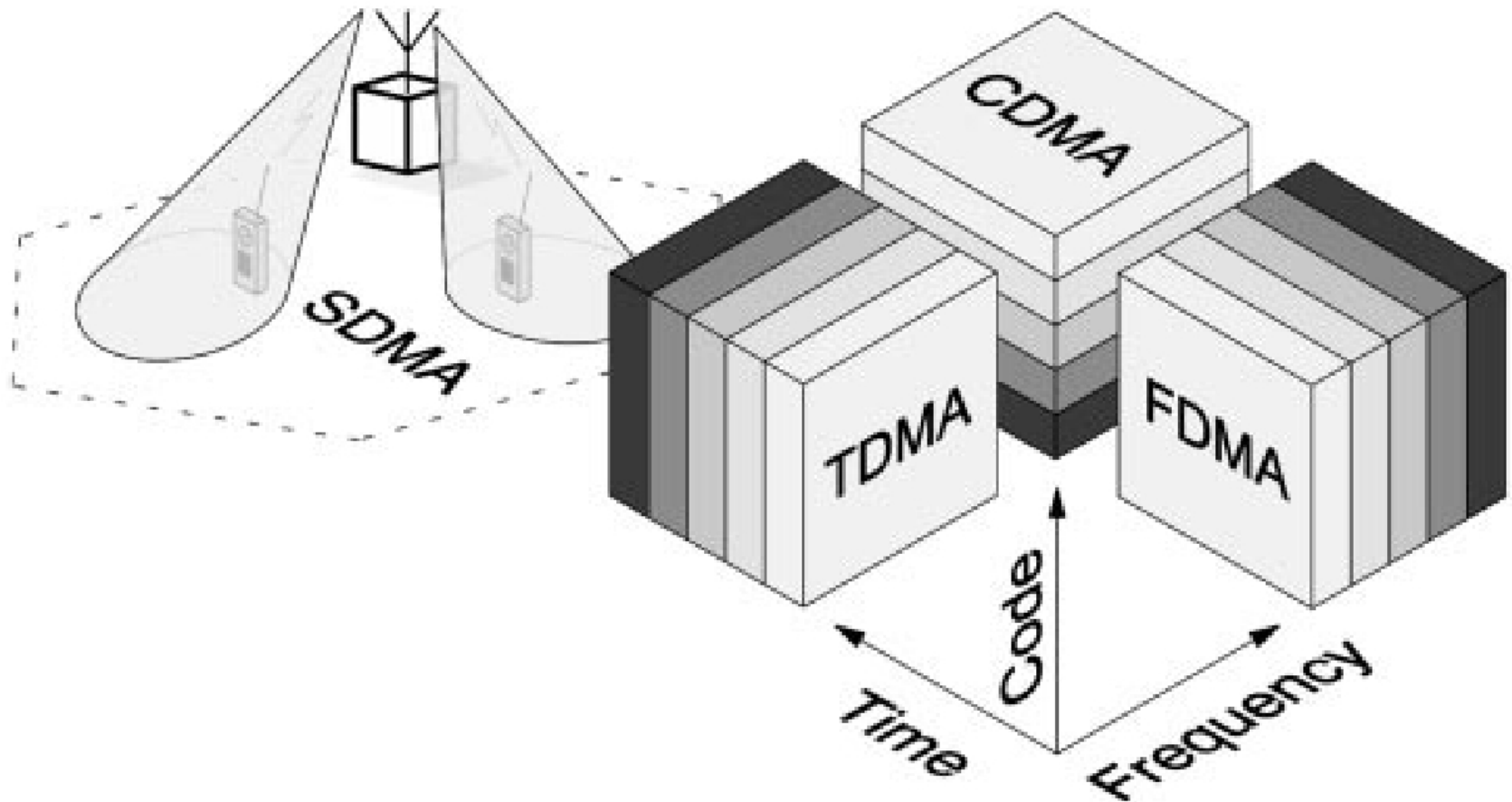
Space Division Multiple Access (SDMA)

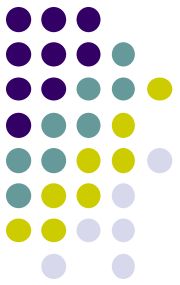


- Space division multiple access (SDMA) controls the radiated energy for each user in space.
- SDMA serves different users by using spot beam antennas.
- Sectorized antennas may be thought of as a primitive application of SDMA,
- In the future, adaptive antennas will be used to simultaneously steer energy in the direction of many users
 - It appear to be best suited for TDMA and CDMA base station architectures.



Multiple access procedures



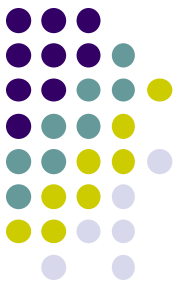


SDMA - Array antennas

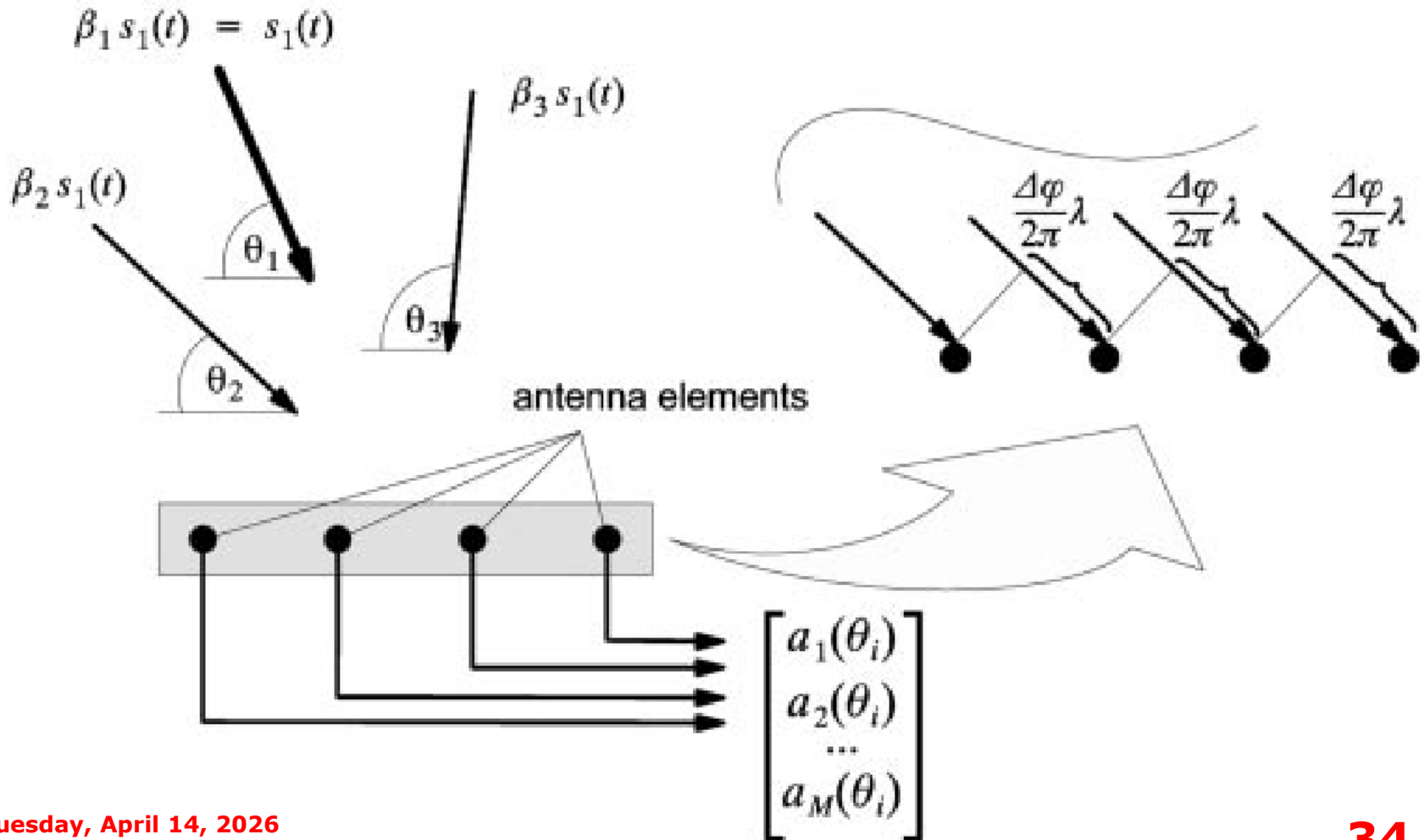
- A perfect adaptive antenna system would be able to track individual multipath components for each user and combine them in an optimal manner to collect all of the available signal energy from each user.
- The perfect adaptive antenna system is not feasible since it requires infinitely large antennas.

Array antennas

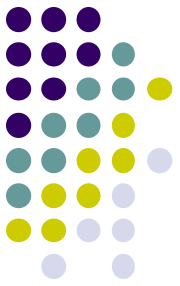
- gains might be achieved using reasonably sized arrays with moderate directivities.



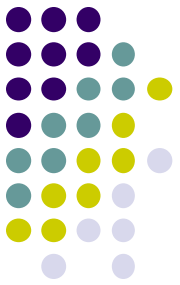
Multipath signal at an antenna array



Space Division Multiple Access (SDMA)

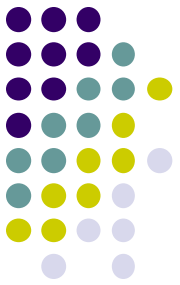


- Multipath propagation
 - frequency-selective fading phenomena
 - the spatial fanning out of signals
 - the received signal to be a summation signal
- The directions of incidence of these multipath components could be distributed arbitrarily at the receiver
- Space division multiple access (SDMA)
- SDMA controls the radiated energy for each user in space.



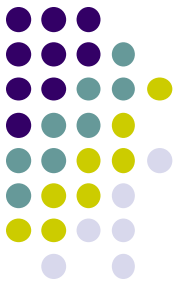
RANDOM ACCESS

- In random access or contention methods,
 - No station is superior to another station and none is assigned the control over another.
 - No station permits, or does not permit, another station to send.
 - At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
 - This decision depends on the state of the medium(idle or busy)



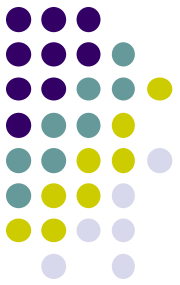
Random Access Protocols

- When node has packet to send
 - transmit at full channel data rate R .
 - no *a priori* coordination among nodes
- Two or more transmitting nodes → “collision”,
- **Random access MAC protocol** specifies:
 - how to detect collisions
 - how to recover from collisions (e.g., via delayed retransmissions)



Random Access Protocols:

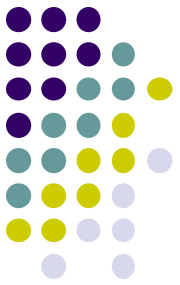
- ALOHA
 - Pure Aloha
 - Slotted ALOHA
- CSMA (Carrier Sense Multiple Access)
- CSMA/CD (CSMA with Collision Detection)
- CSMA/CA (CSMA with Collision Avoidance)



Random Access... Cont'd

■ ALOHA

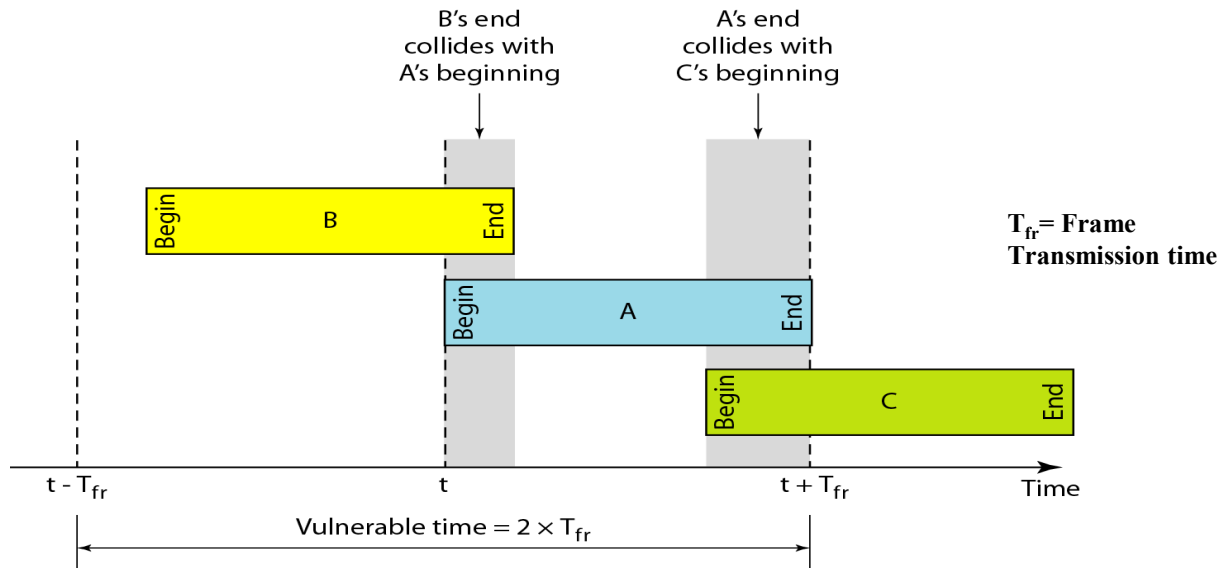
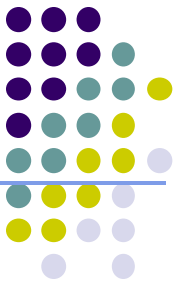
- Developed in the 1970s for a packet radio network by Hawaii University.
- Whenever a station has a data, it transmits. Sender finds out whether transmission was successful or experienced a collision by listening to the broadcast from the destination station. Sender retransmits after some random time if there is a collision.
- ALOHA is the father of multiple access protocols.
 - Pure ALOHA
 - Slotted ALOHA



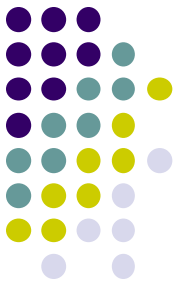
Pure ALOHA

- All frames from any station are of **fixed length (L bits)**
 - Stations transmit at equal transmission time (all stations produce frames with equal frame lengths).
 - A station that has data can transmit at any time
 - After transmitting a frame, the sender waits for an acknowledgment for an amount of time (time out) equal to the maximum round-trip propagation delay
 - Time out= $2 * t_{prop}$
 - If no ACK was received, sender assumes that the frame or ACK has been destroyed and resends that frame after it waits for a random amount of time
 - If station fails to receive an ACK after repeated transmissions, it gives up.
 - Channel utilization or efficiency or Throughput is the percentage of the transmitted frames that arrive successfully (without collisions)

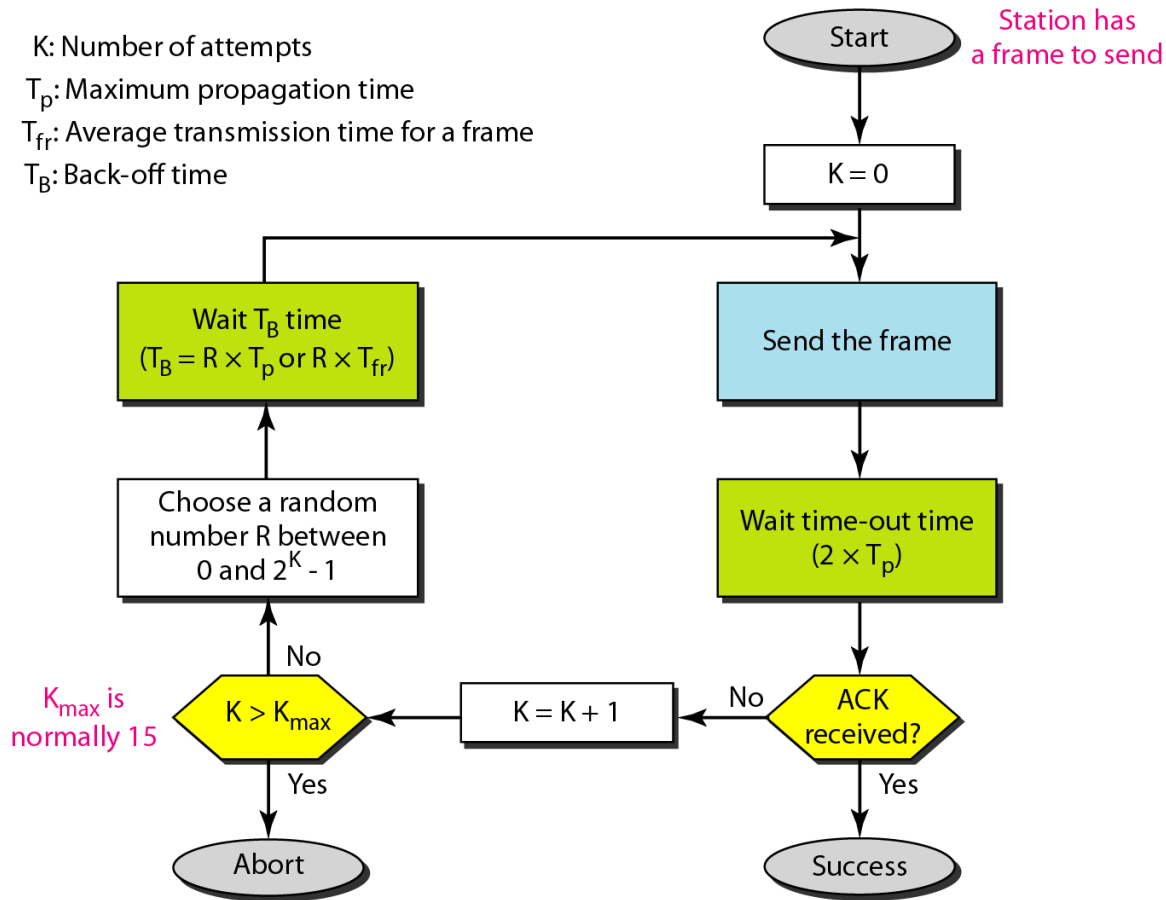
Critical time for pure ALOHA protocol



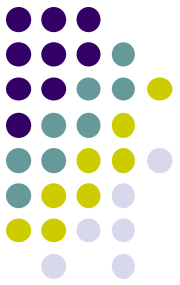
- If the frame transmission time is T_{fr} , then the vulnerable time is $= 2 T_{fr}$.
 - This means no station should send during the T_{fr} before this station starts transmission and no station should start sending during the T -sec period that the current station is sending.



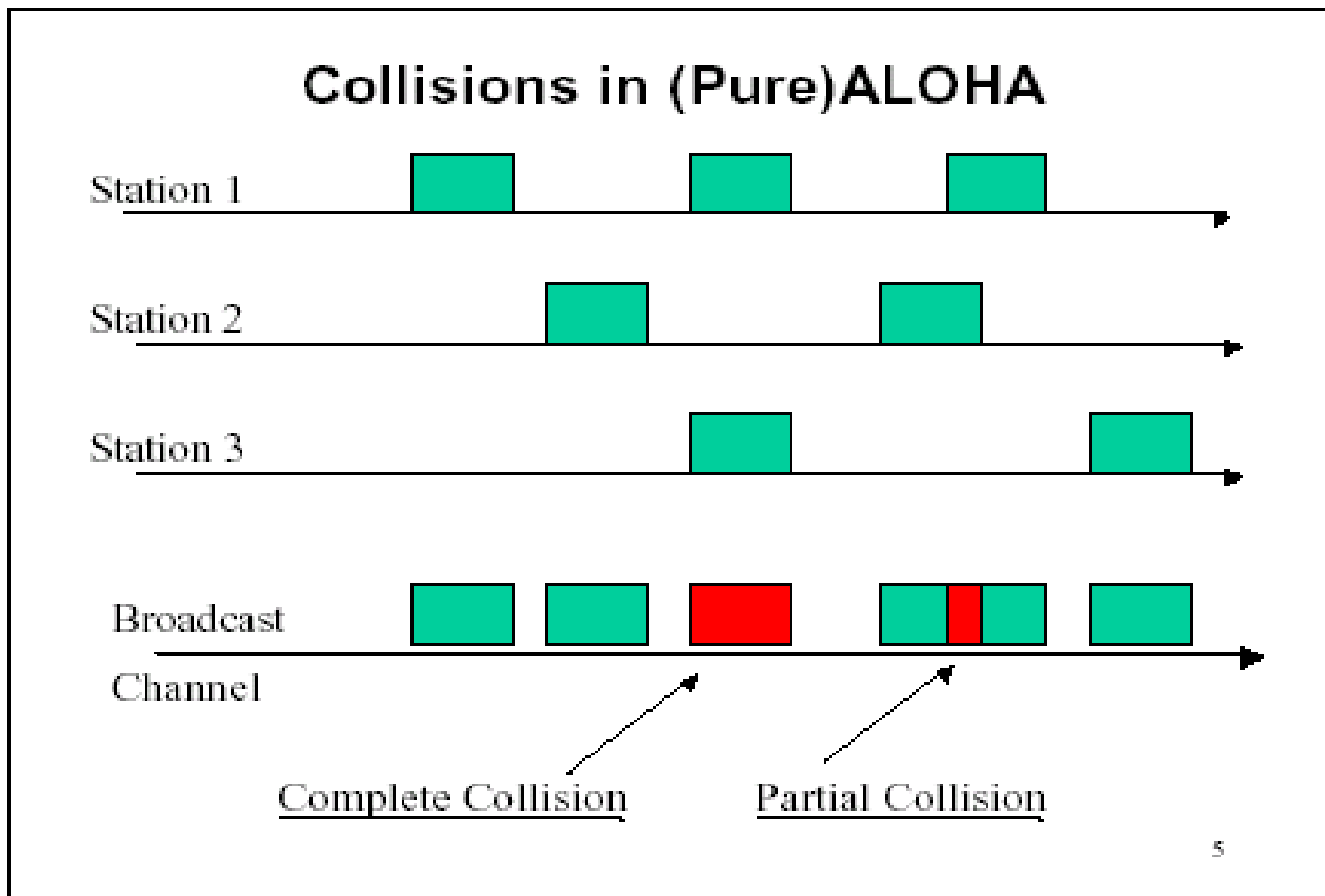
Procedure for pure ALOHA protocol

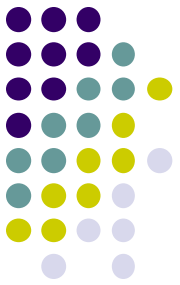


Pure ALOHA



In pure ALOHA, frames are transmitted at completely **random** times.



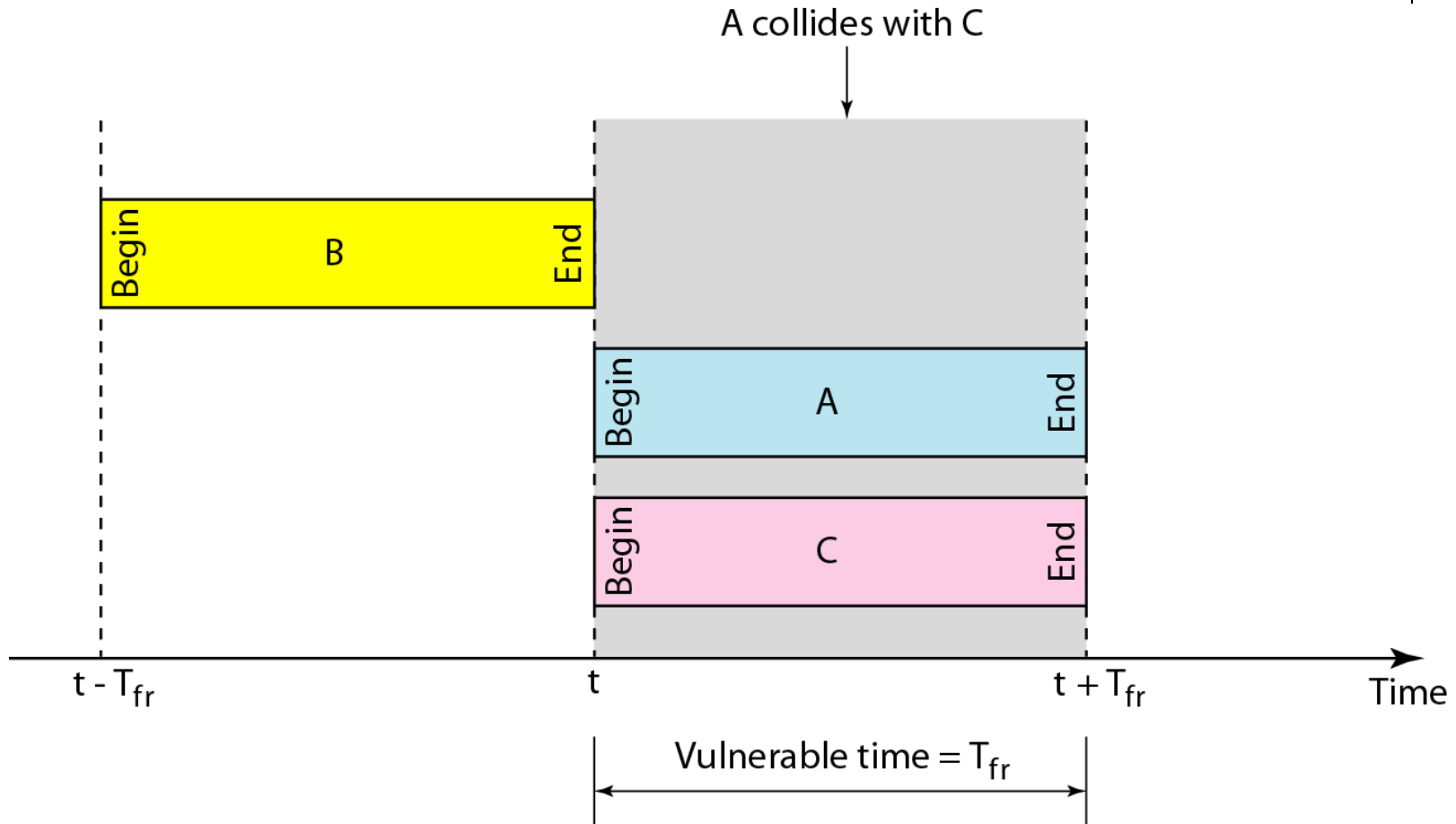
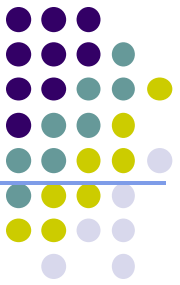


Slotted ALOHA

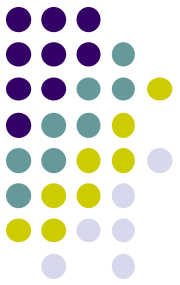
■ Slotted ALOHA

- Time is divided into slots equal to a frame transmission time (T_{fr})
- A station can transmit at the beginning of a slot only.
- If a station misses the beginning of a slot, it has to wait until the beginning of the next time slot.
- A central clock or station informs all stations about the start of a each slot
- **Improvement:** Time is slotted and a packet can only be transmitted at the beginning of one slot. Thus, it can reduce the collision duration.

In danger time for slotted ALOHA protocol

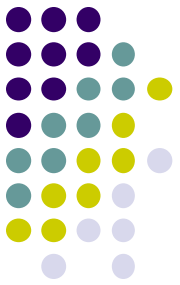


ALOHA

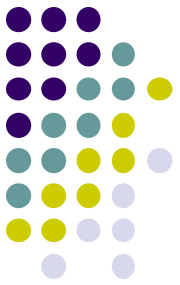


- Advantage of ALOHA protocols
 - A node that has frames to be transmitted can **transmit continuously** at the **full rate of channel (R bps)** if it is the only node with frames
 - Simple to be implemented
 - No master station is needed to control the medium
- Disadvantage
 - If (M) nodes want to transmit, many collisions can occur and the rate allocated for each node will **not be on average R/M bps**
 - This causes low channel utilization

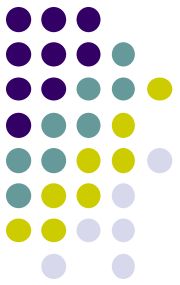
Comparison of ALOHA



BASIS FOR COMPARISON	Pure ALOHA	Slotted ALOHA
Frame Transmission	The user can transmit the data frame whenever the station has the data to be transmitted.	The user has to wait till the next time slot start, to transmit the data frame.
Time	In Pure ALOHA the time is continuous.	In Slotted ALOHA the time is discrete.
Throughput	The maximum throughput occurs at $G = 1/2$ which is 18%	The maximum throughput occurs at $G = 1$ which is 37%.
Danger Time	$= 2 T_{fr}$	$= T_{fr}$



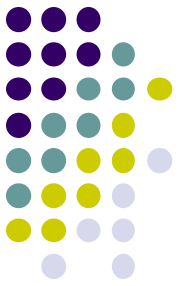
Random Access – Carrier Sense Multiple Access (CSMA)



Carrier Sense Multiple Access (CSMA)

- A user wishing to transmit first listens to the medium to see if another transmission is in progress (carrier sense)
- Based on the fact that in LAN propagation time is very small
 - If a frame was sent by a station, All stations knows immediatel so they can wait before start sending
 - A station with frames to be sent, should sense the medium for the presence of another transmission (carrier) before it starts its own transmission
 - This can reduce the possibility of collision but it cannot eliminate it.
- Collision can only happen when more than one station begin transmitting within a short time (the propagation time period)

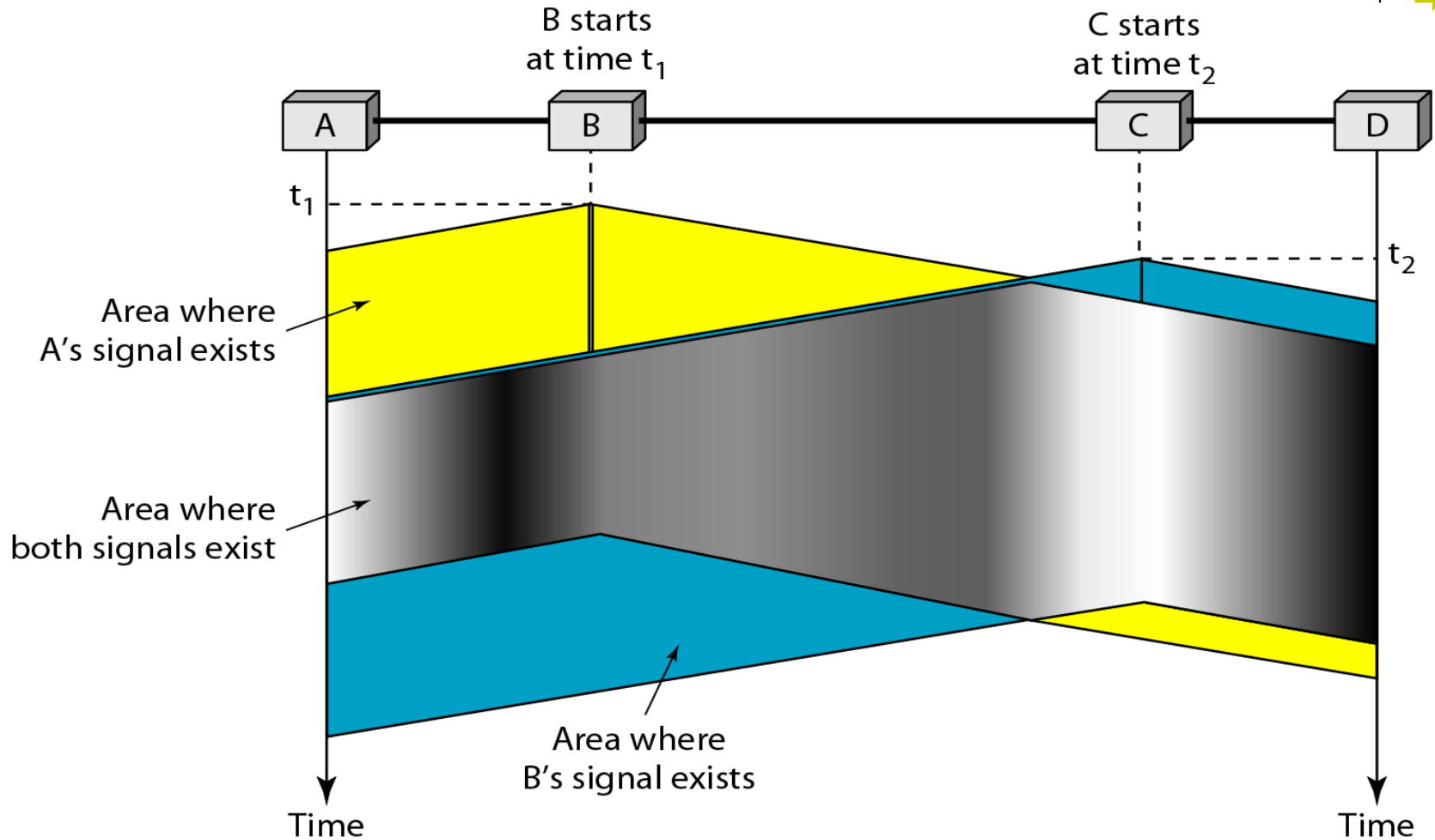
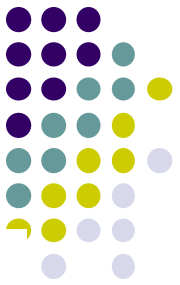
CSMA (Carrier Sense Multiple Access)



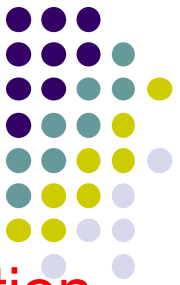
- CSMA: listen before transmit:
- If channel sensed idle: transmit entire frame
- If channel sensed busy, defer transmission

Human analogy: don't interrupt others!

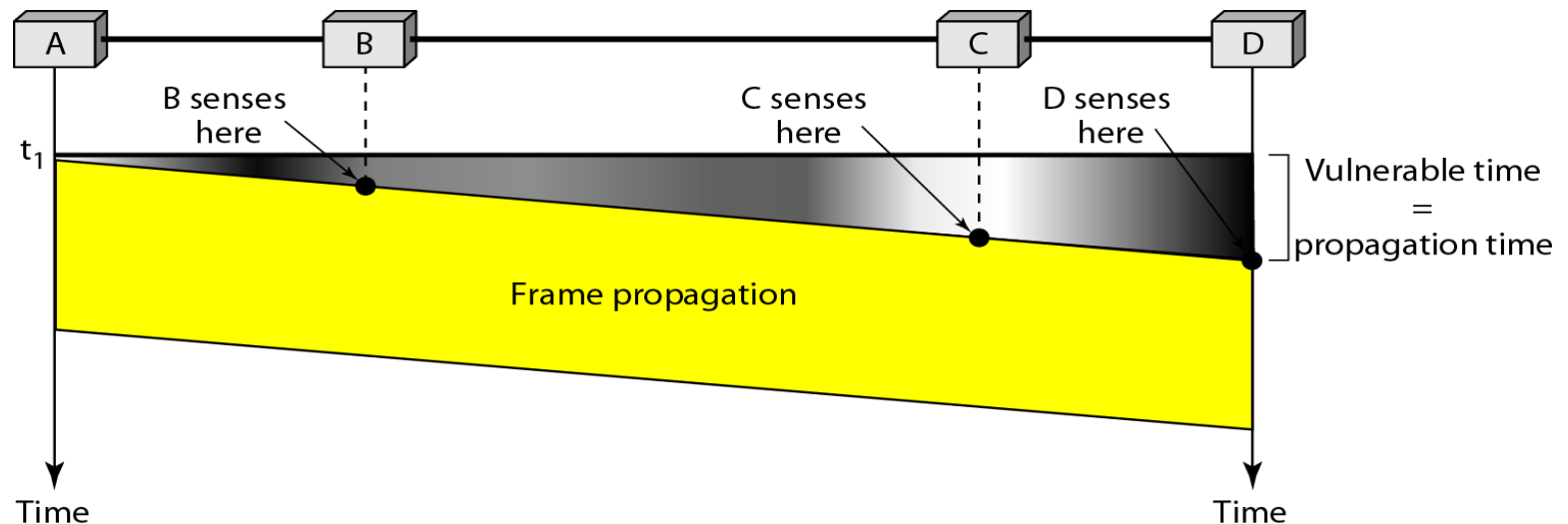
Space/time model of the collision in CSMA

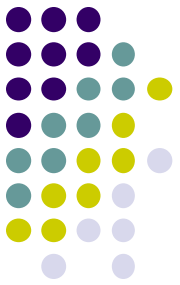


Vulnerable time in CSMA



- Vulnerable time for CSMA is the maximum **propagation time**
 - The longer the propagation delay, the worse the performance of the protocol because of the above case.





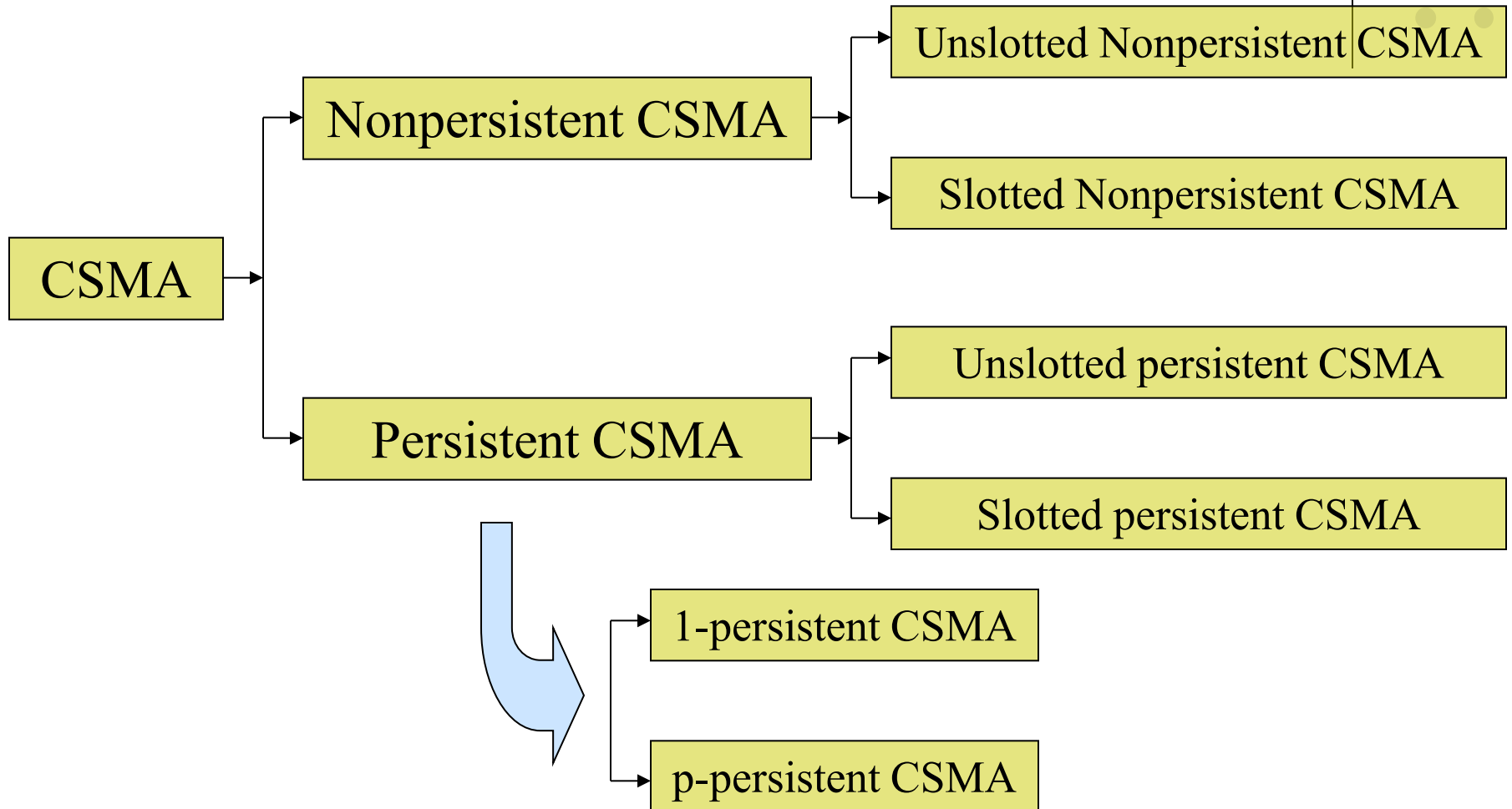
Types of CSMA Protocols

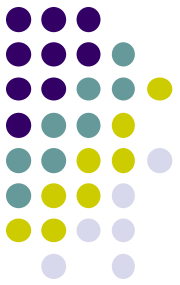
Different CSMA protocols that determine:

- What a station should do when the medium is idle?
- What a station should do when the medium is busy?

1. Non-Persistent CSMA
2. 1-Persistent CSMA
3. p-Persistent CSMA

Types of CSMA





CSMA/CD (Collision Detection)

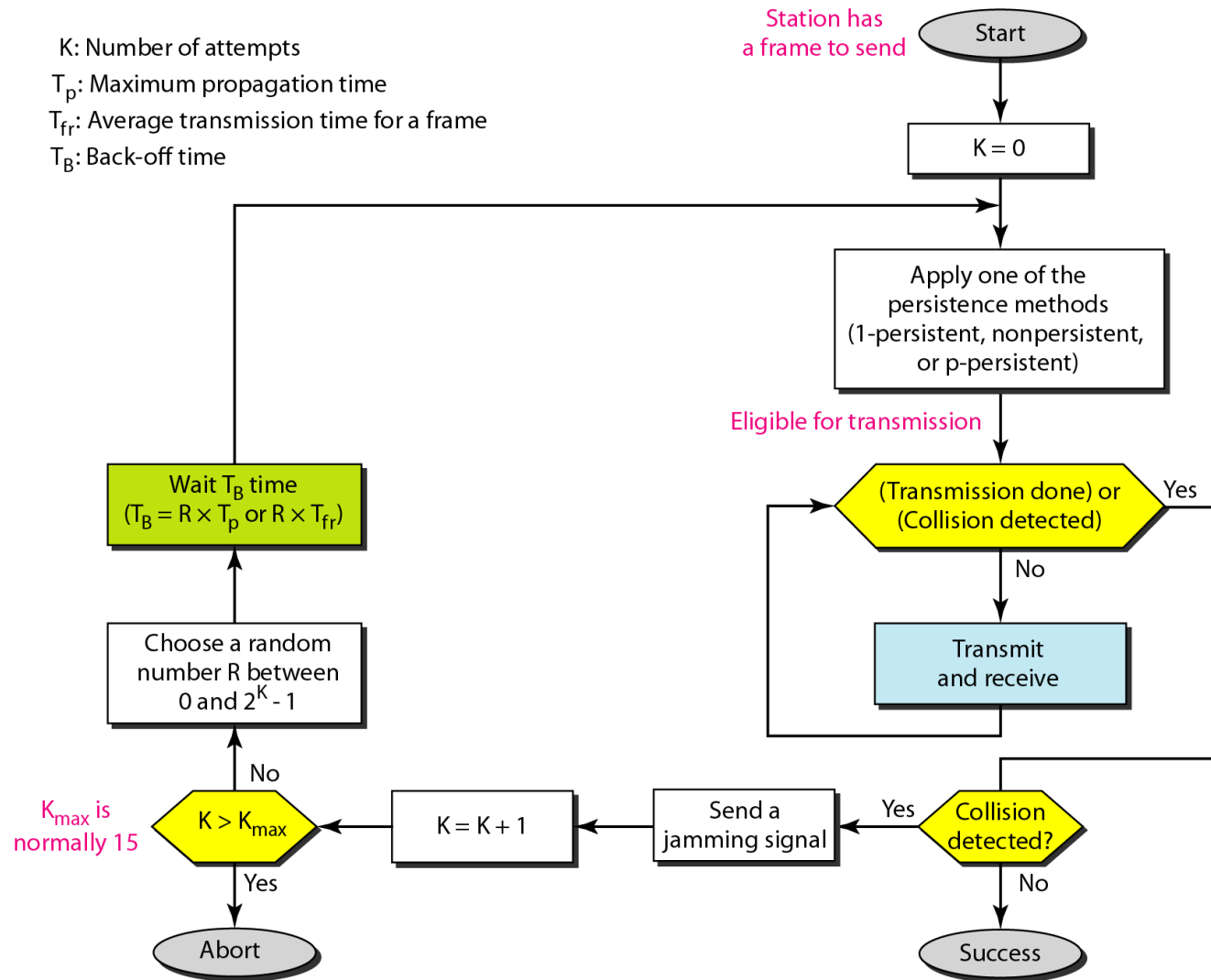
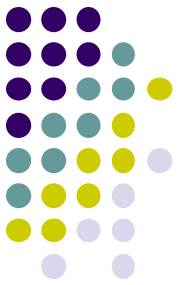
- CSMA (all previous methods) has an inefficiency:
 - If a collision has occurred, the channel is unstable until colliding packets have been fully transmitted
- CSMA/CD (Carrier Sense Multiple Access with Collision Detection) overcomes this as follows:
 - While transmitting, the sender is listening to medium for collisions.
 - Sender stops transmission if collision has occurred reducing channel wastage .
 - Wait random time before you attempt to resend.
 - CSMA/CD is Widely used for bus topology LANs (IEEE 802.3, Ethernet).

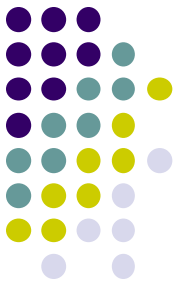


CSMA/CD (CSMA with Collision Detection)

- In CSMA, if 2 terminals begin sending packet at the same time, each will transmit its complete packet (although collision is taking place).
- Wasting medium for an entire packet time.
- CSMA/CD
 - Step 1: If the medium is idle, transmit
 - Step 2: If the medium is busy, continue to listen until the channel is idle then transmit
 - Step 3: If a collision is detected during transmission, cease transmitting
 - Step 4: Wait a random amount of time and repeats the same algorithm

Flow diagram for the CSMA/CD



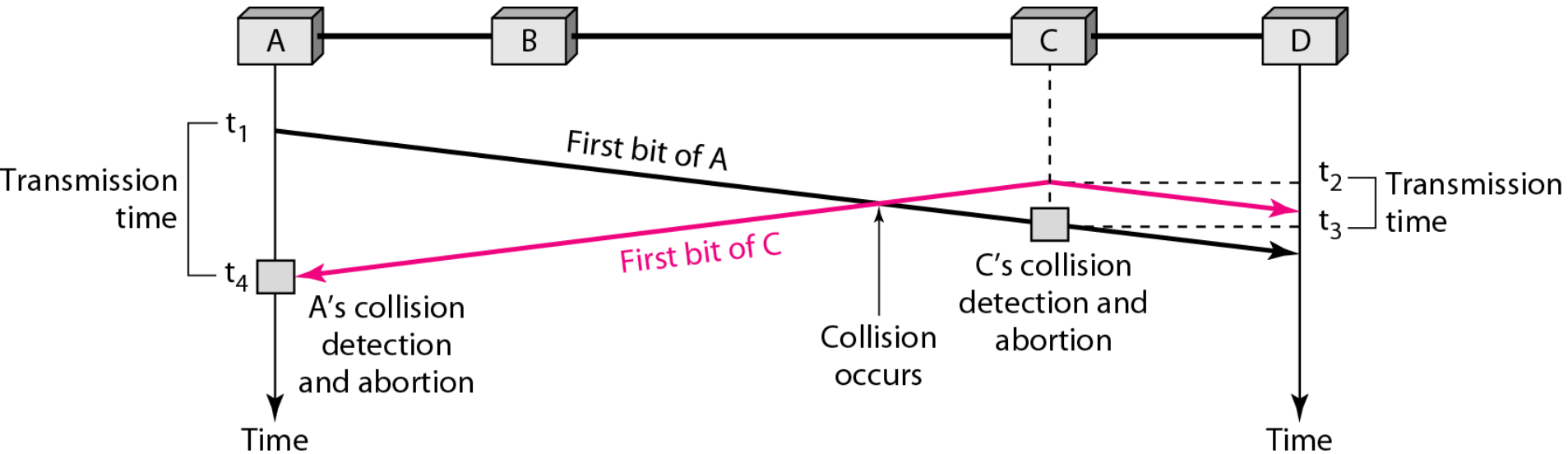
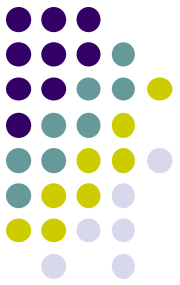


How does a node detect collision?

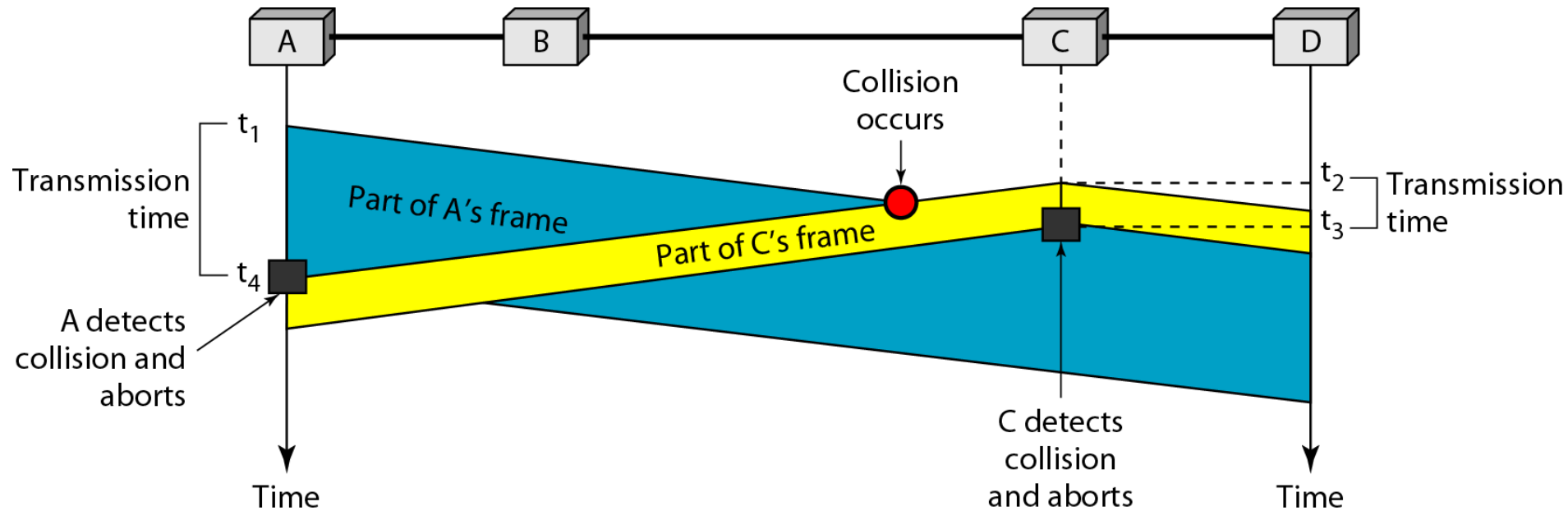
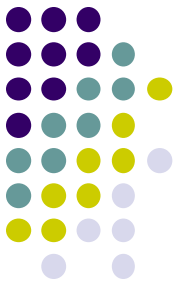
- Transceiver: A node monitors the media while transmitting. If the observed power is more than transmitted power of its own signal, it means collision occurred

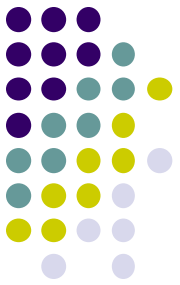


Collision of the first bit in CSMA/CD



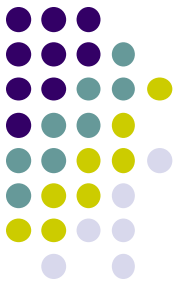
Collision and abortion in CSMA/CD





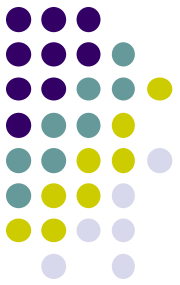
CSMA/CA (CSMA with collision Avoidance)

- All terminals listen to the same medium as CSMA/CD.
 - Terminal ready to transmit senses the medium.
 - If medium is busy it waits until the end of current transmission.
 - It again waits for an additional predetermined time period DIFS (Distributed inter frame Space).
 - Then picks up a random number of slots (the initial value of backoff counter) within a contention window to wait before transmitting its frame.
 - If there are transmissions by other terminals during this time period (backoff time), the terminal freezes its counter.
 - It resumes count down after other terminals finish transmission + DIFS. The terminal can start its transmission when the counter reaches to zero.



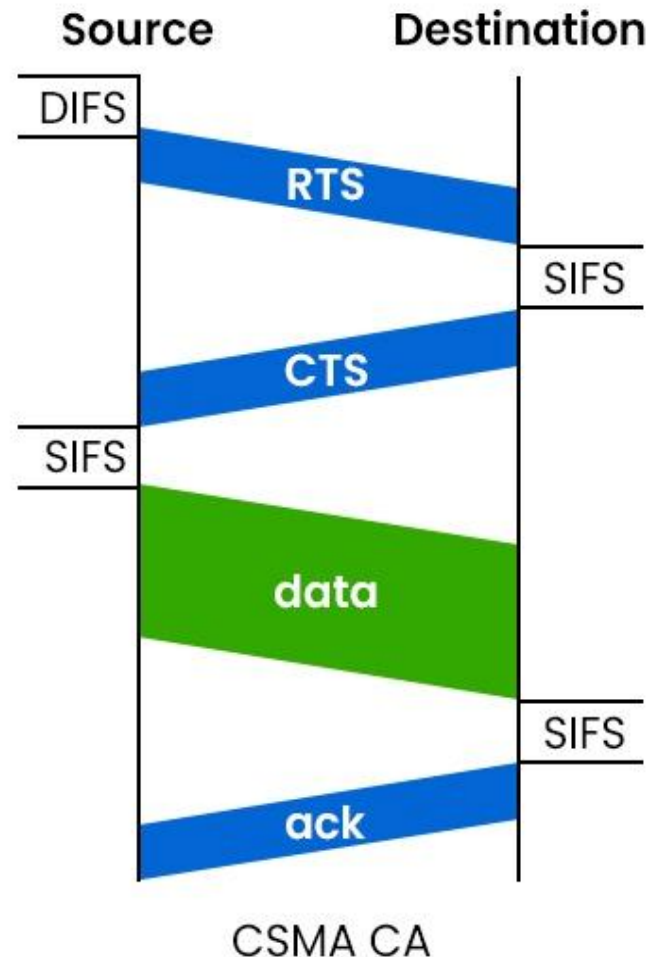
How Does CSMA/CA Work?

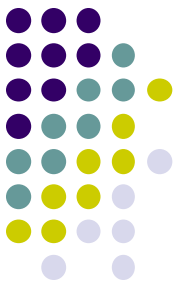
- Wireless Local Area Networks(WLAN) use a protocol called Carrier Sense, Multiple Access/Collision Avoidance (CSMA/CA).
- Collisions cannot be detected in wireless media as they can in Ethernet.
- In a WLAN, a device cannot simultaneously transmit or receive data. It either does one thing or it doesn't. As a result, it is unable to recognize when two frames collide.



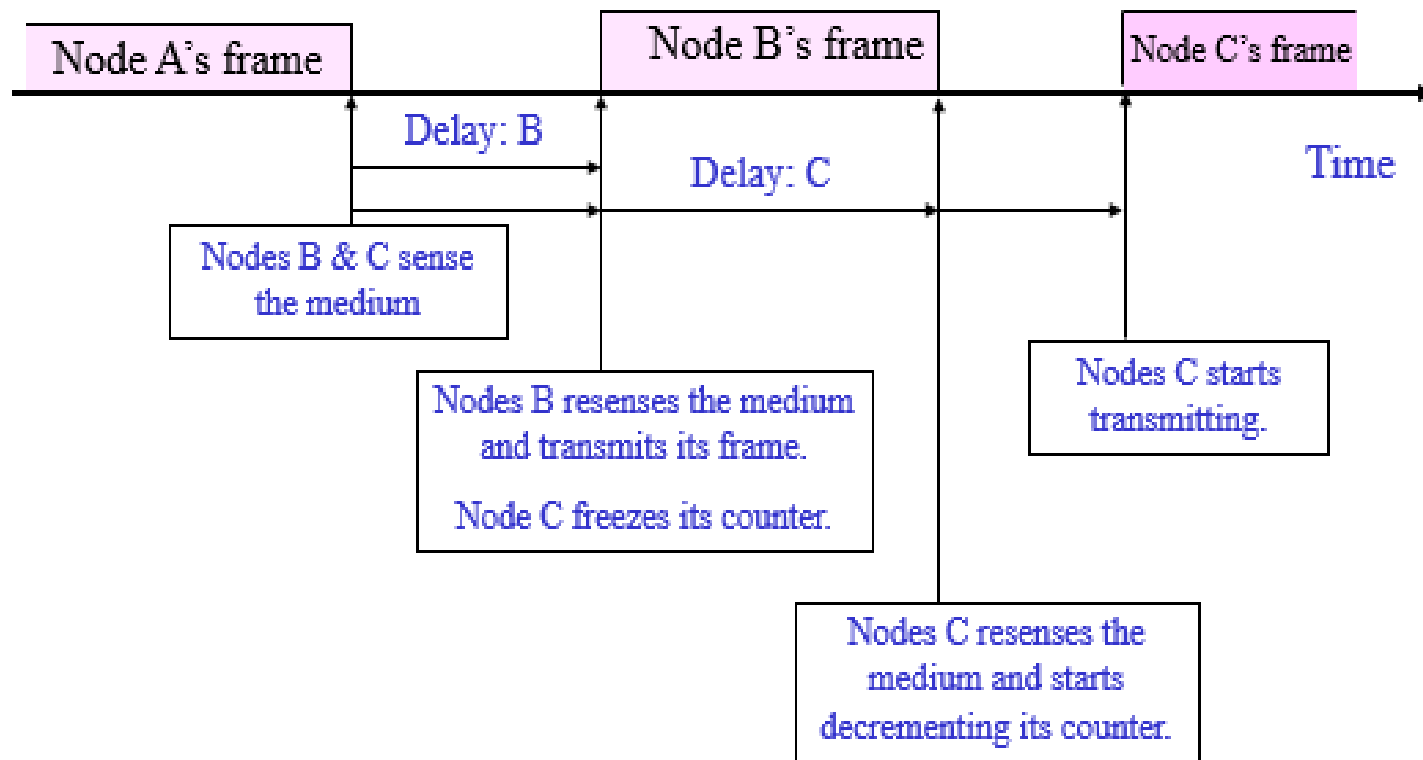
How Does CSMA/CA Work?... Cont'd

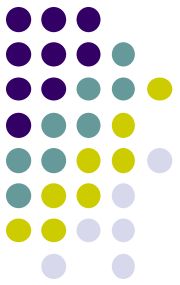
- Figure shows how CSMA/CA works.



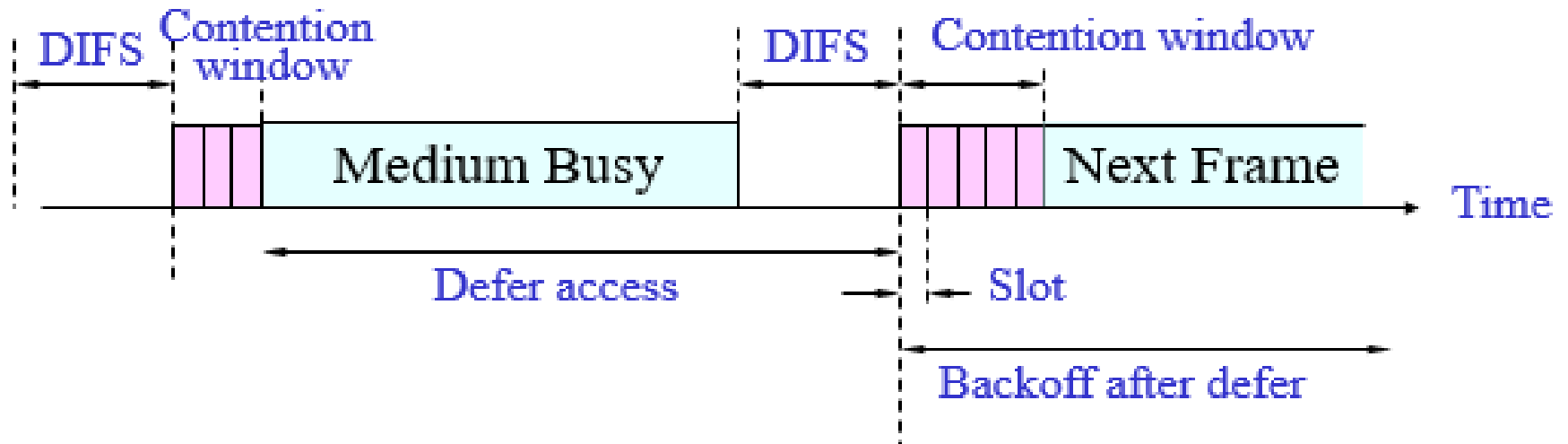


CSMA/CA ... cont'd

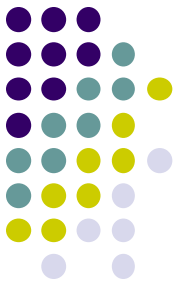




CSMA/CA Explained

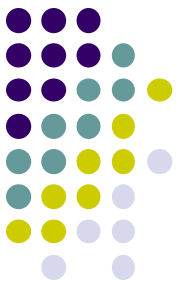


DIFS – Distributed Inter Frame Spacing

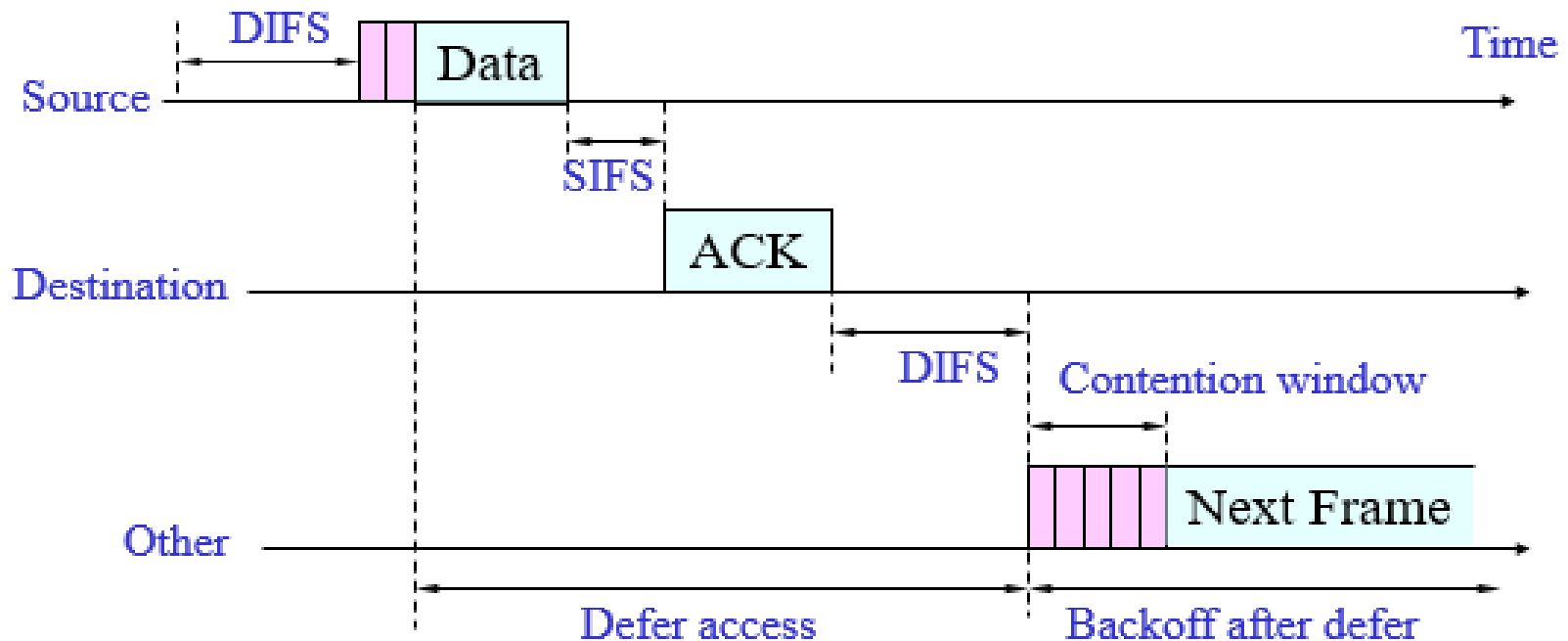


CSMA/CA with ACK

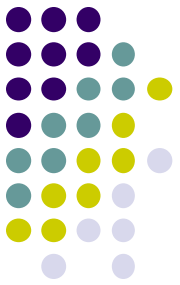
- Immediate Acknowledgements from receiver upon reception of data frame without any need for sensing the medium.
- ACK frame transmitted after time interval SIFS (Short Inter-Frame Space) ($SIFS < DIFS$)
- Receiver transmits ACK without sensing the medium.
- If ACK is lost, retransmission done.



CSMA/CA/ACK

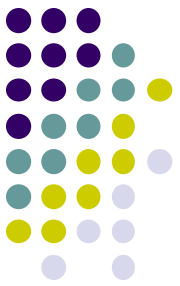


SIFS – Short Inter Frame Spacing

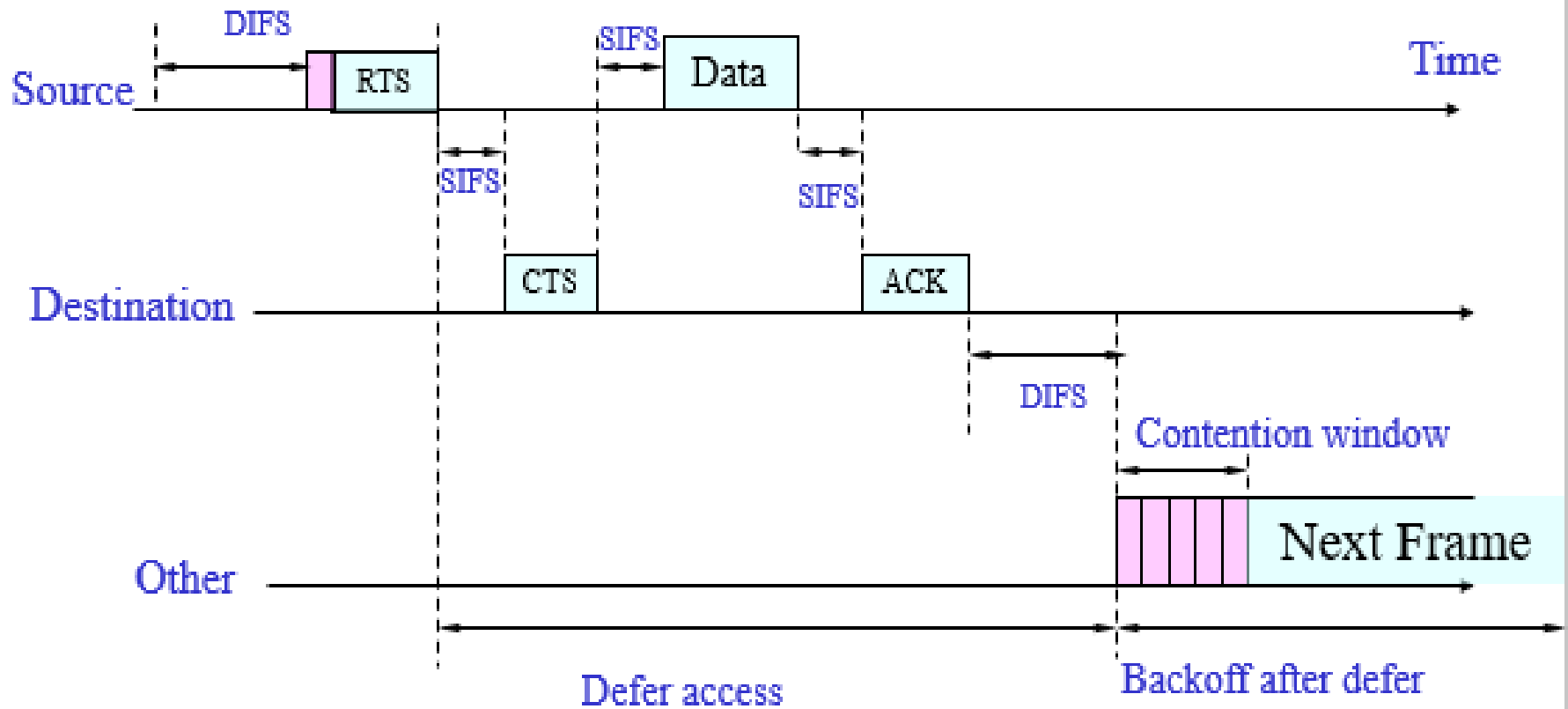


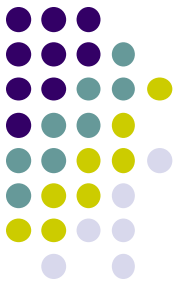
CSMA/CA with RTS/CTS

- Transmitter sends an RTS (request to send) after medium has been idle for time interval more than DIFS.
- Receiver responds with CTS (clear to send) after medium has been idle for SIFS.
- Then Data is exchanged.
- RTS/CTS is used for reserving channel for data transmission so that the collision can only occur in control message.



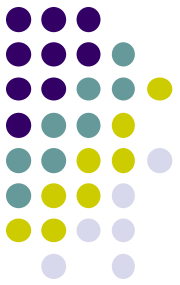
CSMA/CA with RTS/CTS (Cont'd)





Note

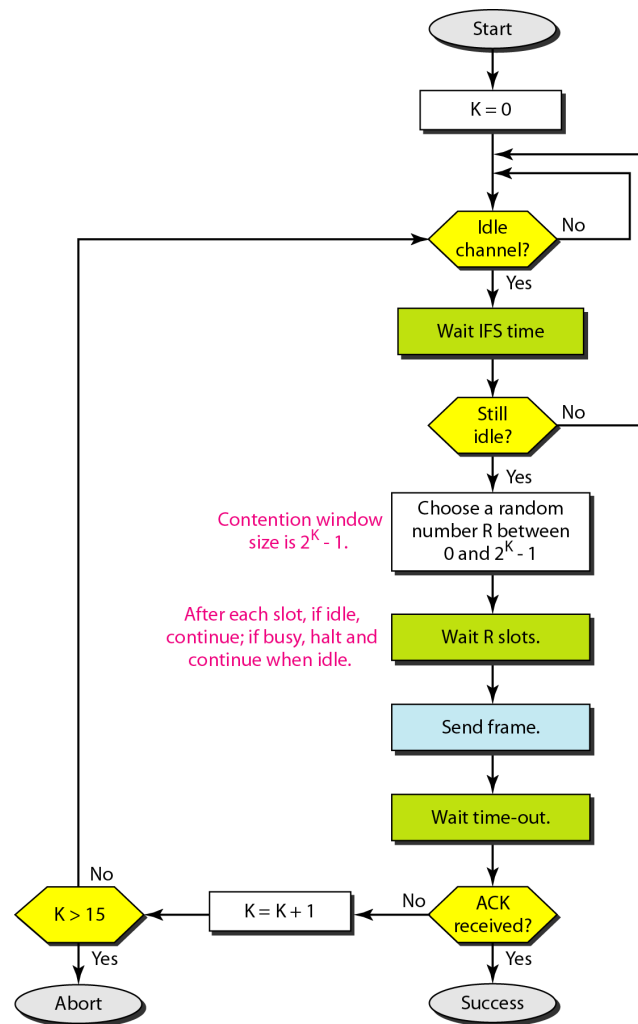
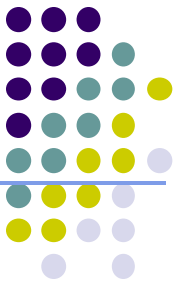
In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.

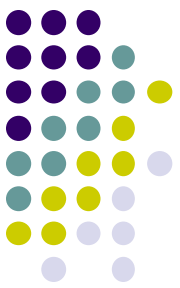


Note

In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

Flow diagram for CSMA/CA

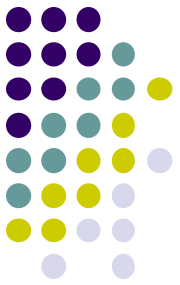




CSMA/CD Vs CSMA/CA

Factors	CSMA CD	CSMA CA
Collison Handling	Detects collisions after they occur and stops transmission.	Avoids collisions before they occur by sensing the channel.
Network Type	Used mostly in wired networks where collision detection is possible.	Used mostly in wireless networks where collision detection is not possible.
Efficiency	Reduces the recovery time after a collision and is more efficient than CSMA CA.	Minimizes the possibility of collision and is less efficient than CSMA CD.
Data Frame Transmission	Resends the data frame whenever a conflict occurs.	Sends an intent to transmit before data transmission.
IEEE Standard	Used in 802.3 standard (Ethernet)	Used in 802.11 standard (Wi-Fi)
Similarity to simple CSMA	More efficient than simple CSMA.	Similar to simple CSMA.

Exercises



1. Why did the 2G cellular systems shift from analog FDMA to digital TDMA and CDMA?
2. Name three standards using TDMA/TDD as their access method?
3. What are the advantages of the CDMA access techniques?
4. What is the difference between performance evaluation of voice oriented fixed assignment and data oriented random access methods ?
5. In radio ALOHA network, how does a terminal learn that its packet is collided?
6. Explain the difference between carrier-sending mechanisms in the wired and wireless channels